



Invarune
by NimeshBuild

SECURITY CONTROLBOOK / SCANNER 0.15.0

AI Agent & MCP Security

Evidence for agent security.

Agents, MCP servers, skills and built images.
Source-linked controls for systems that reason and act.

66

CONTROLS

132

ACCEPTANCE CHECKS

47

STATIC RULES

NSA + CISA / CSA / MITRE / NIST

OWASP / MCP / CIS / standards + research benchmarks

CONTROLS RESEARCH 19 SEP / DETECTOR UPDATE 23 SEP 2026

github.com/nimeshbuilds/invarune

NAVIGATION

Contents

Click a row or use the PDF bookmarks to move through the complete source-linked controlbook.

Reading guide and assurance labels	3
Source landscape	4
Security review workflow	5
Coverage: Execution and data flow / MCP identity	6
Coverage: Prompt injection / Secrets and data handling	7
Coverage: Skills and malicious-tool indicators / Built images and supply chain	8
Coverage: Operations and control effectiveness	9
Scan configuration and optional review	10
CLI examples and selected scans	11
Transparent metrics and score calculation	12
Human review and evidence-bound import	13
Controlled security analyst	14
Benchmarks and attack suites	15
Governance	18
Identity and authorization	21
MCP protocol and tools	26
Agent behavior and context	31
Execution and application security	35
Data and privacy	39
Supply chain	42
Operations and resilience	45
Security validation	48
47-rule clickable inventory	53
Every scan: algorithm, limits, fixes and source mapping	57
Full primary-source directory	104

01 / READING GUIDE

From code patterns to assurance

The Invarune catalog by NimeshBuild, with evidence requirements and a traceable source for every control.

Use this book to review agents, MCP clients and servers, gateways, tools, retrieval, memory, and the infrastructure that grants them authority. It is a synthesized engineering checklist, not a reproduction of every external framework or a certification claim.

FIND THE CONTROL YOU NEED

Governance	06 controls	18
Identity and authorization	09 controls	21
MCP protocol and tools	10 controls	26
Agent behavior and context	07 controls	31
Execution and application security	07 controls	35
Data and privacy	06 controls	39
Supply chain	06 controls	42
Operations and resilience	06 controls	45
Security validation	09 controls	48

Controlled security analyst: page 14

Static-rule index: page 53 / **Full source directory:** page 104

Read the labels carefully. A source link shows provenance or thematic alignment. It does not mean that every acceptance check is quoted from that source, or that a rule satisfies an entire external requirement. The applicability and limitations of each source appear in the directory.

This controlbook is a reference, not a bound scan report. For editable findings and fresh-scan import, use the report created by `invscan --report`, then pass the edited artifact with `--review-report`.

02 / SOURCE LANDSCAPE

Different sources. Different jobs.

Combine guidance, control catalogs, protocol obligations, deployment baselines, and measured attack behavior.

NSA + CISA

GOVERNMENT GUIDANCE

Agent adoption, secure deployment, data protection, lifecycle practices, and incident readiness.

CSA

AI + CLOUD CONTROLS

AI Controls Matrix, MAESTRO threat modeling, cloud shared responsibility, and supplier evidence.

NIST

RISK + SECURE DEVELOPMENT

AI RMF, GenAI Profile, adversarial ML taxonomy, SSDF and its AI-specific profile.

MITRE ATLAS

THREAT KNOWLEDGE BASE

Attacker techniques and attack paths. Use to build scenarios; it is not an assurance score.

OWASP

RISK + VERIFICATION CATALOGS

Agentic and LLM risks, MCP guidance, and AI security verification requirements.

MCP

PROTOCOL SPECIFICATION

Version-sensitive transport, authorization, tool, sampling, elicitation, and trust requirements.

CIS

HARDENING BENCHMARKS

Agent/MCP companion guidance and configuration baselines for applicable server and deployment layers.

ISO / OpenSSF / SLSA

MANAGEMENT + SOFTWARE INTEGRITY

Governance context, project security practices, and build provenance. Applicability varies by artifact.

AIUC + research suites

ASSURANCE + EXPERIMENTS

Additional agent-control context and executable attacks under explicit harness assumptions.

This is a broad, curated landscape verified for this edition. No finite snapshot covers every publication or future benchmark. External catalogs retain their own scope, version, terms, and assessment procedures.

03 / VALIDATION METHOD

Make every decision reviewable

Keep deterministic signals, runtime evidence, human assessment, and LLM opinions separate.

1

Scope the system

Record agent/server identity, tools, model and gateway versions, MCP revision, data classes, deployment, and who can influence each input. Include downstream side effects and credentials.

2

Run the static scan

The CLI offers 47 deterministic rules; `--scans` selects rules or controls. Keep findings, evidence, severity, confidence, remediation, exceptions, and coverage gaps. An absent pattern is not a passed control.

3

Test the effective boundary

Use isolated deployments, synthetic credentials, explicit canaries, and safe effect sinks. Verify identities, tenant boundaries, approvals, egress, cancellation, quotas, and audit records.

4

Measure attacks and useful work

Pin the benchmark, model, tools, settings, and attack budget. Report unauthorized actions and data leakage alongside benign task completion, false blocking, sample size, and repeated attempts.

5

Record a human disposition

Retain owner, scope, evidence and review date. Report decisions are justified, disabled, note, `needs_runtime_validation` or `needs_human_review`. Justified and disabled are exceptions, never validated passes.

Optional controlled security analyst

When enabled in full mode, every active selected check is queued through deterministic retrieval, bounded requests, strict schemas and exact-quote checks. Missing answers remain unreviewed. The LLM remains nondeterministic; runtime and owner verification stay open. Static findings and the severity gate remain intact. Review contract: page 14.

METHOD COVERAGE

What each layer can establish

Real catalog counts and explicit blind spots. Mapping is not accuracy, compliance, or percent secure.

Controls with static rule mappings		30
Controls without static rule mappings		36

Execution and data flow

DETERMINISTIC: Python AST checks propagate selected registered-tool inputs and same-file arguments/returns, retain aliases and exact literal constraints, and check selected execution/file/network sinks. JavaScript tokens support narrow registerTool callbacks and direct-return named/arrow wrappers. Explicit world-writable chmod modes are checked.

OPTIONAL ANALYST: A model can form a trust-boundary hypothesis, request bounded captured source ranges to inspect callers or guards, seek counterevidence and explain what the supplied code still cannot establish.

CAN MISS OR MISCLASSIFY: Cross-file flows, unknown decorators/methods, complex JavaScript wrappers, dynamic imports, reflection, generated code and runtime values can hide a risk. Recognized resource/binding failures remain gaps. A dangerous API can be intentional and correctly constrained.

VALIDATE ELSEWHERE: Exercise actual entry points with authorized negative tests; verify input provenance, execution privileges and sandbox boundaries.

MCP identity, authorization and tools

DETERMINISTIC: Selected configuration/code signals flag insecure transport, explicit authentication bypass, token passthrough, broad tool grants and unpinned server launch packages.

OPTIONAL ANALYST: A model can review supplied authorization logic, tool descriptions and trust-boundary evidence for missing assumptions or inconsistent checks.

CAN MISS OR MISCLASSIFY: Configuration may differ from deployment. External gateways, identity providers, token audience checks and tenant boundaries may be absent from retrieved evidence.

VALIDATE ELSEWHERE: Test expired/wrong-audience tokens, cross-tenant access, per-tool authorization, consent binding and the deployed MCP transport.

METHOD COVERAGE

What each layer can establish

Real catalog counts and explicit blind spots. Mapping is not accuracy, compliance, or percent secure.

Prompt injection, retrieval and memory

DETERMINISTIC: Selected patterns identify external content placed in privileged model messages and explicit approval/sandbox bypass. This is not an attack simulation.

OPTIONAL ANALYST: A model can reason about untrusted-content handling, persistent memory, tool-output influence and missing approval boundaries in the supplied excerpts.

CAN MISS OR MISCLASSIFY: Adaptive attacks, multi-turn interactions and tool chains depend on runtime behavior. Retrieval can omit the relevant code; a model can overlook an attack or be influenced by hostile evidence.

VALIDATE ELSEWHERE: Run isolated adversarial agent/MCP evaluations with attack-success and legitimate-task measurements, plus human review of consequential actions.

Secrets and data handling

DETERMINISTIC: Selected literal credentials, plausible private-key material, credential-bearing URLs, logging patterns and browser credential exposure are flagged and redacted best-effort.

OPTIONAL ANALYST: A model can identify sensitive data flows and retention concerns when the supplied evidence explains their context.

CAN MISS OR MISCLASSIFY: Encoded, split or runtime-fetched credentials can be missed; labels/test values can resemble secrets. Neither layer establishes whether a credential is live or redaction is complete.

VALIDATE ELSEWHERE: Verify secret stores, credential rotation, logging/retention policy, egress destinations and actual data-access permissions.

METHOD COVERAGE

What each layer can establish

Real catalog counts and explicit blind spots. Mapping is not accuracy, compliance, or percent secure.

Skills and malicious-tool indicators

DETERMINISTIC: Bounded skill/tool/schema descriptions are checked for instruction hijacking, sensitive-data transfer, covert or approval-bypassing actions. Selected Spanish/French/German overrides and paired authority claims supplement English predicates. Python read-only tool annotations are compared with recognized direct write witnesses.

OPTIONAL ANALYST: A model can review intent, indirect social engineering, skill reference context and tool behavior visible in retrieved evidence, including cases where syntax or supported patterns are inconclusive.

CAN MISS OR MISCLASSIFY: A pattern is risk evidence, not proof of malicious authorship. Obfuscation, remote references, external packages, omitted context and runtime behavior can evade either layer.

VALIDATE ELSEWHERE: Inspect publisher provenance and the exact installed skill/tool; test consequential actions in an authorized isolated environment and verify effective permissions.

Built images and supply chain

DETERMINISTIC: Image archives are reconstructed without starting containers; supported packaged source, metadata, retained credentials, stored permissions and package inventories are inspected.

OPTIONAL ANALYST: A model can interpret supplied image/source evidence, flag missing deployment context and propose verification work.

CAN MISS OR MISCLASSIFY: Compiled logic, live deployment overrides, signature trust and dependency CVEs are not assessed. Package inventory and archive consistency do not establish safety or publisher identity.

VALIDATE ELSEWHERE: Use vulnerability/advisory analysis, provenance/signature verification and deployment testing alongside the static image assessment.

METHOD COVERAGE

What each layer can establish

Real catalog counts and explicit blind spots. Mapping is not accuracy, compliance, or percent secure.

Operations and control effectiveness

DETERMINISTIC: The catalog retains governance, monitoring, recovery, approval and isolation checks even where no deterministic rule maps to them.

OPTIONAL ANALYST: Full review queues every active acceptance check and can identify evidence gaps or propose runtime/human validation; it can cite exact submitted source text.

CAN MISS OR MISCLASSIFY: A valid citation proves that a quote exists, not that the model's interpretation is correct. Policies, external services, incident procedures and production outcomes may be unavailable.

VALIDATE ELSEWHERE: Record owner-reviewed evidence, operational exercises and runtime results. A written justification remains an accepted exception, never an automatically validated pass.

CONFIGURATION

Choose the evidence, then the limits

The default scan is offline and deterministic. Optional review and PDF dependencies are selected explicitly.

TARGET

Source directory, local Docker/Podman image reference, or Docker-save/OCI archive. No target program or container is started.

BOUNDS + EXCLUSIONS

Declare file/byte/entry/image limits and exclusions. Unsupported syntax, truncated evidence and exhausted budgets remain visible gaps.

DETERMINISTIC OUTPUT

Without `--report/--output`, results stay in the terminal. `--report [DIR]` or `--output DIR` saves HTML, Markdown, JSON and SARIF without LLM credentials; `--pdf` adds an interactive PDF.

OPTIONAL ANALYST

Select an API/custom gateway configuration or the official Codex, Claude Code or Grok Build CLI. Review can be full-control or findings-only, with explicit call/time/evidence bounds.

USER DISPOSITIONS

`--review-config` applies explicit rule/control/check exceptions. `--review-report` imports edited report fields into a fresh scan. Neither is automatically loaded from the target.

EXIT + ASSURANCE

The findings threshold is a policy gate, not a security grade. Operational gaps and required human/runtime validation remain incomplete. Justified is distinct from pass.

CLI FIELD GUIDE

Choose exactly what to inspect

The `invscan` command works without an LLM. Inventory, explanations and control mappings are bundled for offline use.

DISCOVER**`invscan --list-scans`**

List all deterministic rules and control review plans. Use `--catalog-format json` for automation and `--help-topic all` for the complete CLI guide.

UNDERSTAND**`invscan --explain-scan AI043`**

Show the predicate, algorithm, applicability, limits, remediation, mapped controls and source relationships for a rule. Control IDs work too.

TERMINAL ONLY**`invscan ./my-agent`**

Inspect supported source/configuration, agent instructions and skills. Without a report flag, no report directory is written.

SELECT SCANS**`invscan ./my-agent --scans AI001,AI043 --scans MCP-03`**

Rule IDs select their patterns and mapped control review; control IDs select their mapped rules/checks. Shared parsing and integrity gaps remain visible.

WRITE REPORTS**`invscan ./my-agent --report ./results --pdf`**

Save HTML, Markdown, JSON, SARIF and optional fillable PDF. PDF needs the pdf extra; deterministic output needs no model credentials.

IMAGE + OPTIONAL REVIEW**`invscan --image-archive ./agent.tar --judge-cli claude`**

Inspect a supported archive without starting the container. Optional full review uses bounded evidence; missing packaged source and unresolved runtime behavior stay unknown.

A control with no mapped detector remains explicitly unvalidated. `--scans` does not silently substitute all rules. `--judge-mode findings` narrows optional review to finding triage; full mode also reviews active selected checks, including zero-match and inconclusive evidence.

INTERPRETATION

Metrics, not a security grade

A percentage can describe review coverage. It cannot establish the chance that an agent, server or skill is secure.

DETERMINISTIC ATTENTION

Count open findings by severity; urgent = critical + high. No arbitrary severity weights or probability of compromise are assigned. The finding gate uses the configured severity threshold.

PARTIAL MAPPING REACH

$100 \times \text{active selected controls with at least one active selected mapped rule} / \text{active selected controls}$. This measures available partial rule coverage, not completed tests or passes.

OPTIONAL AI ANSWER COVERAGE

$100 \times \text{unique active selected acceptance checks with a valid received model answer} / \text{active selected acceptance checks}$. Concerns and explicit unknowns count as answers; synthesized omissions do not.

EMPTY SCOPE + EXCEPTIONS

An empty denominator is null (not applicable), never 100%. Justified and disabled checks/rules are excluded from active denominators without positive or negative credit. Observed evidence and reasons remain visible.

AI DOES NOT ERASE EVIDENCE

AI-disabled means zero answered checks. Enabling AI cannot lower recorded severity, erase a deterministic finding or change the deterministic severity gate. Runtime/human validation stays open; model code support is advisory.

Worked example using this catalog

With every control selected and no exceptions: $30/66 = 45.45\%$ partial mapping reach. That is not a security score. If 3 of 6 active checks receive valid model answers, answer coverage is 50% even when every answer says runtime validation is required.

Read the final result in order

1. Scope completion and coverage gaps.
2. Open critical/high concerns and fixes.
3. User exceptions and retained evidence.
4. Optional advice, actual answered checks and unknowns.
5. Required runtime/human validation.

A clean pattern scan is never a validated pass.

HUMAN REVIEW

From a report to an auditable decision

Review fields carry evidence bindings. Edited reports cannot replay a target, command, gateway or model automatically.

SCAN

01

Choose source/image and configuration. Collect static evidence and a bound review workspace.

REVIEW

02

Inspect the finding or check. Run required human/runtime validation in the authorized environment.

RECORD

03

Choose a decision. Record reason, reviewer, reviewed-at date and a reference to supporting evidence.

SAVE

04

Save HTML/Markdown/JSON/SARIF review data or a compatible fillable PDF with current appearances.

IMPORT

05

Run a fresh explicit target scan with `--review-report`. Origin, source, catalog and item bindings are rechecked.

RESOLVE

06

Stale or invalid edits stay visible for re-review. Only valid, applicable decisions affect the result.

Decision semantics: justified/disabled require an accountable reviewer and reason; they are excluded from active denominators, never counted as passes. Notes do not waive findings. Runtime/human-validation decisions keep work outstanding. Coverage gaps cannot be waived. PDF canonical fields, widgets and appearance text must agree; ambiguous or flattened forms need JSON review instead.

03 / VALIDATION METHOD / CONTINUED

Controlled security analyst

Deterministic boundaries around nondeterministic expertise. Source review produces advice; it does not establish deployed security.

01 Route every active selected check

Static rules provide partial evidence. Full mode queues active selected checks, including controls with no matches or no mapped rule. With the full catalog and no exceptions, this is 66 controls / 132 checks. An empty scan never closes a control.

02 Collect evidence deterministically

Read only unchanged files in the scan manifest. Check file hashes and path confinement, exclude credential files, redact excerpts, and select bounded context using fixed control terms and finding locations.

03 Run the bounded analyst

Use the configured model or custom JSON gateway. Fixed batches and call, size, and time budgets constrain review. The model may request exact ranges from captured file IDs with a risk and counterevidence rationale. The controller authorizes snapshot-only reads; no arbitrary paths, URLs, tools or target execution.

04 Validate the response contract

Require known control/check IDs, allowed statuses, and exact evidence quotes for support, gaps, or proposed non-applicability. Derive file and line metadata from submitted excerpts. Reject malformed or fabricated output.

05 Keep gaps visible and auditable

Record every check, explanation, citation, verification step, and request receipt. Omitted checks and exhausted budgets remain unreviewed. The first failed batch stops further calls; static results are preserved.

Advisory outcomes

Code support / potential gap / runtime validation needed / human review needed / insufficient evidence / non-applicability proposed. Code support cannot complete manual or dynamic controls. Quote validation proves presence in an excerpt, not the model's interpretation.

Default full-catalog review envelope

Up to 36 shared requests, plus finding triage; batches of 6 and 2 optional evidence rounds. Reserve a conclusion call for every remaining batch. Evidence: 200 files / 2 MB; 240 seed excerpts, 120,000 shared characters. Scheduling: 600 seconds, not a hard deadline. Record served/denied range requests and unresolved checks. Use `--analyst-investigation-rounds 0` for seed-only review.

04 / EXECUTABLE EVALUATION

Test behavior, not just code

Research benchmark results apply to a defined harness and threat model. None was executed to produce this handbook.

AgentDojo

[BENCH-AGENTDOJO / NeurIPS 2024; living repository](#)

Indirect prompt injection and legitimate task utility in tool workflows.

Use with care: Simulated suites and evolving API; adapt to production tools and policies. Not executed here.

[Open the primary source](#)

InjecAgent

[BENCH-INJECAGENT / 2024 research; living repository](#)

Malicious tool-result scenarios for tool-integrated agents.

Use with care: Fixed cases are not an adaptive attacker or production permission test. Not executed here.

[Open the primary source](#)

Agent Security Bench (ASB)

[BENCH-ASB / ICLR 2025; living repository](#)

Multiple agent scenarios and attack/defense evaluation methods.

Use with care: Match harness, agent, and tools; published scores do not transfer automatically. Not executed here.

[Open the primary source](#)

04 / EXECUTABLE EVALUATION

Test behavior, not just code

Research benchmark results apply to a defined harness and threat model. None was executed to produce this handbook.

MCPSecBench

[BENCH-MCPSECBENCH / 2025 research; living repository](#)

MCP security playground and attack scenarios across protocol surfaces.

Use with care: Environment/client-specific research workflows; not arbitrary-repository auditing or current-spec conformance. Not executed here.

[Open the primary source](#)

MCP Security Bench (MSB)

[BENCH-MSB / Paper v2; ICLR 2026; living repository](#)

Real MCP tool attacks across planning, invocation, and response handling with a security/utility trade-off metric.

Use with care: Distinct from MCPSecBench; selected tools, domains, agents, and code revisions bound results. Not executed here.

[Open the primary source](#)

MCP-SafetyBench

[BENCH-MCP-SAFETY / Paper v2; living repository](#)

Multi-turn, cross-server safety evaluation using real MCP servers in five domains and 20 attack types.

Use with care: Service state, harness assumptions, and task selection affect reproducibility; not protocol authorization certification. Not executed here.

[Open the primary source](#)

04 / EXECUTABLE EVALUATION

Test behavior, not just code

Research benchmark results apply to a defined harness and threat model. None was executed to produce this handbook.

AgentDyn

[BENCH-AGENTDYN / Paper v3; living repository](#)

Dynamic open-ended tasks and helpful third-party instructions expose over-blocking by injection defenses.

Use with care: Extends AgentDojo and inherits harness assumptions; avoid double-counting identical scenarios. Not executed here.

[Open the primary source](#)

AgentHarm

[BENCH-AGENTHARM / ICLR 2025; living Inspect Evals implementation](#)

Harmful multi-step user requests and jailbreaks with corresponding benign tasks.

Use with care: Different threat model from indirect injection. Grading can use LLMs; public subsets, scoring revisions, and safety/security-use license apply. Not executed here.

[Open the primary source](#)

BIPIA

[BENCH-BIPIA / 2023 initial paper; living repository](#)

Indirect prompt injection in external email, web, table, summarization, and code content.

Use with care: External-content robustness, not full agent/MCP transport security; some datasets require separate licensed acquisition. Not executed here.

[Open the primary source](#)

05 / CONTROL CATALOG

Governance

GOV-01 - GOV-02 / Check applicability, retain evidence, and resolve exceptions.

GOV-01

MANUAL VALIDATION

Inventory every agent, MCP server, tool, and model

- ☐ Record owner, deployment, model/version, MCP transport/version, exposed tools, data classes, and external endpoints.
- ☐ Reconcile approved inventory with deployed configurations; investigate unregistered agents and servers.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-RMF](#) / [CSA-AICM](#) / [CSA-AISMM](#) / [CIS-AGENTS-2026](#) / [CIS-MCP-2026](#)

GOV-02

MANUAL VALIDATION

Model trust boundaries and attack paths

- ☐ Diagram user, model, memory, tool, server, gateway, and downstream service boundaries with their credentials.
- ☐ Identify who controls each input and the highest-impact action reachable if that input is malicious.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [MITRE-ATLAS](#) / [NIST-AML](#) / [CSA-MAESTRO](#) / [MITRE-ATLAS-202609](#)

05 / CONTROL CATALOG

Governance

GOV-03 - GOV-04 / Check applicability, retain evidence, and resolve exceptions.

GOV-03

MANUAL VALIDATION

Define authorized use and accountable owners

- ☐ Assign an owner to each autonomous action and document permitted purposes, forbidden outcomes, and escalation routes.
- ☐ Record impact, reversibility, approval requirements, and accepted residual risk before production use.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-RMF](#) / [JOINT-AGENTIC](#) / [CSA-AICM](#) / [CSA-SCOPING](#) / [CSA-AISMM](#) / [AIUC1-Q3-2026](#)

GOV-04

MANUAL VALIDATION

Maintain evidence and risk exceptions

- ☐ Give each control an owner, evidence link, validation date, result, and next review date.
- ☐ Time-limit exceptions and require a compensating control; distinguish untested behavior from demonstrated failure.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-GENAI](#) / [NIST-SSDF](#) / [CSA-AICM](#) / [CSA-AISMM](#) / [NIST-SSDF-AI](#)

05 / CONTROL CATALOG

Governance

GOV-05 - GOV-06 / Check applicability, retain evidence, and resolve exceptions.

GOV-05

MANUAL VALIDATION

Reassess changes in autonomy and integrations

- ☐ Review security impact when adding a model, tool, server, data source, skill, or broader permission.
- ☐ Require deployment evidence for the complete configured system; a model-only score is insufficient.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NCSC-SECURE-AI](#) / [CSA-AICM](#) / [CSA-SCOPING](#)

GOV-06

MANUAL VALIDATION

Assign shared security responsibilities across AI providers

- ☐ For each model, gateway, orchestrator, MCP service, and cloud provider, document which party implements each applicable safeguard and which customer configuration it depends on.
- ☐ Obtain current supplier evidence for inherited safeguards, identify unowned gaps, and record reassessment triggers in the service review.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [CSA-AICM](#) / [CSA-CCM](#)

05 / CONTROL CATALOG

Identity and authorization

AUTH-01 - AUTH-02 / Check applicability, retain evidence, and resolve exceptions.

AUTH-01

HYBRID VALIDATION

Authenticate protected operations on every request

- ☐ Trace authentication to every protected HTTP entry point, including tool calls, subscriptions, and retries.
- ☐ Verify missing, expired, revoked, or malformed credentials cannot invoke a protected operation.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI026, AI041 (partial)

Source context: [MCP-AUTH](#) / [CSA-AGENT-IAM](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C10](#)

AUTH-02

HYBRID VALIDATION

Authorize the exact action and target

- ☐ Check user/agent identity, tenant, tool, target object, and requested operation immediately before execution.
- ☐ Deny by default; test read-only callers against write tools and object identifiers owned by another user.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI027 (partial)

Source context: [JOINT-AGENTIC](#) / [MCP-TOOLS](#) / [CSA-AICM](#) / [CSA-AGENT-IAM](#) / [CSA-SCOPING](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#) / [OWASP-AISVS-C10](#)

05 / CONTROL CATALOG

Identity and authorization

AUTH-03 - AUTH-04 / Check applicability, retain evidence, and resolve exceptions.

AUTH-03

HYBRID VALIDATION

Validate access-token cryptography and claims

- ☐ Verify signature with trusted keys, allowed algorithms, expected issuer/audience, expiry, and required scopes.
- ☐ Reject unsigned tokens and tokens minted for another service; decoding a JWT alone is not validation.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI017 (partial)

Source context: [MCP-AUTH-SEC](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C10](#)

AUTH-04

HYBRID VALIDATION

Prevent token passthrough and confused deputy use

- ☐ Use audience-bound MCP credentials and separately authorized downstream credentials; never forward arbitrary caller tokens.
- ☐ Ensure a proxy cannot use its broader service identity to perform an action the caller cannot authorize.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI028 (partial)

Source context: [MCP-AUTH](#) / [MCP-SECURITY](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C10](#)

05 / CONTROL CATALOG

Identity and authorization

AUTH-05 - AUTH-06 / Check applicability, retain evidence, and resolve exceptions.

AUTH-05

HYBRID VALIDATION

Constrain credential lifetime and exposure

- ☐ Use short-lived scoped credentials where supported, protect refresh tokens, and validate rotation and revocation.
- ☐ Avoid tokens in query strings, model context, source, child-process arguments, and diagnostic output.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI010, AI011, AI030, AI034 (partial)

Source context: [MCP-AUTH-SEC](#) / [JOINT-DEPLOY](#) / [CSA-AGENT-IAM](#) / [CIS-AGENTS-2026](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#)

AUTH-06

HYBRID VALIDATION

Bind OAuth flows and validate redirects

- ☐ Test PKCE support and S256, exact registered redirects, transaction binding, and authorization-response issuer validation.
- ☐ Reject replayed codes, mismatched issuers, unsafe redirect schemes, and unsolicited callback transactions.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI038 (partial)

Source context: [MCP-AUTH-SEC](#) / [MCP-AUTH](#)

05 / CONTROL CATALOG

Identity and authorization

AUTH-07 - AUTH-08 / Check applicability, retain evidence, and resolve exceptions.

AUTH-07

HYBRID VALIDATION

Secure OAuth discovery and registration

- ☐ Validate discovered metadata and client metadata URLs before fetching; constrain schemes, destinations, redirects, and response sizes.
- ☐ Document trust policy for client registration and prevent discovery from reaching internal metadata or privileged network services.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [MCP-SECURITY](#)

AUTH-08

HYBRID VALIDATION

Bind delegation to identity, scope, and expiry

- ☐ Carry authenticated initiator and delegation identity through multi-agent calls instead of trusting identity fields in text.
- ☐ Prevent agents from granting themselves privileges; limit delegation scope, depth, lifetime, and downstream audiences.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [OWASP-AGENT-CS](#) / [CSA-AGENT-IAM](#) / [NIST-AGENT-IDENTITY-DRAFT](#) / [CIS-AGENTS-2026](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#)

05 / CONTROL CATALOG

Identity and authorization

AUTH-09 - AUTH-09 / Check applicability, retain evidence, and resolve exceptions.

AUTH-09

DYNAMIC VALIDATION

Manage agent identity enrollment and retirement

- ☐ Enroll agents under an accountable sponsor and approved workload identity; verify identity claims before issuing credentials or granting discovery and execution access.
- ☐ Test retirement, sponsor departure, redeployment, and identity compromise; remove stale credentials, cached grants, registrations, and downstream access.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.**Static coverage:** No mapped rule; obtain review or runtime evidence.**Source context:** [CSA-AGENT-IAM](#) / [NIST-AGENT-IDENTITY-DRAFT](#)

ASSESSMENT RECORD

Use this space to record evidence for the control above. A source citation or a clean scan alone is not a validation result.

Owner / review date

Evidence / test run

Result / exception / next action

05 / CONTROL CATALOG

MCP protocol and tools

MCP-01 - MCP-02 / Check applicability, retain evidence, and resolve exceptions.

MCP-01

HYBRID VALIDATION

Validate transport exposure and origin

- ❑ For HTTP, reject invalid Origin values and verify local deployments bind only to intended interfaces.
- ❑ Use TLS for remote protected endpoints; test DNS rebinding and proxy/header behavior in deployment.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI006, AI007, AI008, AI029 (partial)

Source context: [MCP-HTTP](#) / [OWASP-AISVS-C10](#)

MCP-02

HYBRID VALIDATION

Validate tool arguments and results

- ❑ Apply schemas and semantic bounds before executing every tool; reject unknown properties where appropriate.
- ❑ Bound sizes and nesting; validate declared structured output and render errors without leaking secrets.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [MCP-TOOLS](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#) / [OWASP-AISVS-C10](#)

05 / CONTROL CATALOG

MCP protocol and tools

MCP-03 - MCP-04 / Check applicability, retain evidence, and resolve exceptions.

MCP-03

HYBRID VALIDATION

Treat descriptions and annotations as untrusted

- ☐ Inspect descriptions, schemas, resources, icons, and results for instructions that cross tool or user boundaries.
- ☐ Never let readOnlyHint, destructiveHint, or other server claims replace independent authorization and approval policy.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI043, AI044, AI046 (partial)

Source context: [MCP-TOOLS](#) / [OWASP-MCP-CS](#) / [OWASP-AISVS-C10](#)

MCP-04

HYBRID VALIDATION

Detect tool substitution and metadata changes

- ☐ Bind tool approval to verified server identity and the reviewed tool definition or version.
- ☐ Revalidate changes after reconnect/list updates; disambiguate collisions across servers without trusting display names.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI046 (partial)

Source context: [OWASP-MCP10](#) / [OWASP-AISVS-C10](#)

05 / CONTROL CATALOG

MCP protocol and tools

MCP-05 - MCP-06 / Check applicability, retain evidence, and resolve exceptions.

MCP-05

DYNAMIC VALIDATION

Protect state handles and legacy sessions

- ☐ Authorize every state handle against its owner and tenant; enforce expiry, unpredictability, and replay boundaries.
- ☐ For older sessionful protocol versions, test session hijacking and cross-user resumption; a session ID is not authentication.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: AI038 (partial)

Source context: [MCP-TOOLS](#) / [MCP-SECURITY](#)

MCP-06

DYNAMIC VALIDATION

Constrain sampling and returned model context

- ☐ Apply host policy and user control to sampling requests, context sharing, model selection, and associated tool access.
- ☐ Test whether an untrusted server can induce disclosure from unrelated conversations or recursively consume model budget.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [MCP-SAMPLING](#)

05 / CONTROL CATALOG

MCP protocol and tools

MCP-07 - MCP-08 / Check applicability, retain evidence, and resolve exceptions.

MCP-07

HYBRID VALIDATION

Secure elicitation and URL interactions

- ☐ Keep sensitive credential collection out of form-mode elicitation; validate and visibly identify URL destinations.
- ☐ Test cancellation, phishing URLs, unsolicited interactions, and replayed completion state against the selected protocol version.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [MCP-ELICITATION](#)

MCP-08

HYBRID VALIDATION

Enforce filesystem isolation independently of roots

- ☐ Treat declared roots as scoped information, not an operating-system sandbox or complete authorization mechanism.
- ☐ Enforce allowed paths at file access and test traversal, symlinks, alternate encodings, and writes outside the workspace.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI015 (partial)

Source context: [MCP-ROOTS](#)

05 / CONTROL CATALOG

MCP protocol and tools

MCP-09 - MCP-10 / Check applicability, retain evidence, and resolve exceptions.

MCP-09

HYBRID VALIDATION

Constrain local server launch and inherited environment

- ☐ Approve executable and package identity before starting a stdio server; use argument arrays and a minimal environment.
- ☐ Restrict proxy process-spawn APIs and child filesystem/network permissions; separate stdout protocol traffic from logs.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI018, AI031 (partial)

Source context: [MCP-SECURITY](#) / [OWASP-MCP-CS](#) / [OWASP-AISVS-C10](#)

MCP-10

DYNAMIC VALIDATION

Apply version-aware metadata, caching, and cancellation

- ☐ Record supported revisions and test their capability, metadata, header consistency, stream, and cancellation rules.
- ☐ Prevent caches and request continuations from crossing authorization contexts; do not apply legacy handshake assumptions universally.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [MCP-TRANSPORTS](#) / [MCP-HTTP](#)

05 / CONTROL CATALOG

Agent behavior and context

AGT-01 - AGT-02 / Check applicability, retain evidence, and resolve exceptions.

AGT-01

HYBRID VALIDATION

Enforce action policy outside the model

- ☐ Place allow/deny decisions at the execution boundary using trusted policy inputs and constrained tool capabilities.
- ☐ Show that an injected instruction cannot disable policy, choose privileged credentials, or bypass approval.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI027, AI031 (partial)

Source context: [ASD-HARNESS](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#) / [AIUC1-Q3-2026](#)

AGT-02

DYNAMIC VALIDATION

Bind approval to the action executed

- ☐ Present actual recipient, target, arguments, data disclosure, and consequences for high-impact approval.
- ☐ Invalidate approval if arguments or target change; test races, delayed retries, and approval reuse.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: AI031, AI045 (partial)

Source context: [OWASP-AGENT-CS](#) / [CSA-SCOPING](#) / [OWASP-AISVS-C9](#)

05 / CONTROL CATALOG

Agent behavior and context

AGT-03 - AGT-04 / Check applicability, retain evidence, and resolve exceptions.

AGT-03

HYBRID VALIDATION

Separate untrusted content from authoritative instructions

- ☐ Track origin and trust level for web pages, documents, messages, OCR, tool results, and repository instructions.
- ☐ Test direct and indirect goal hijacking; formatting delimiters and prompt warnings alone are not access controls.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI032, AI043 (partial)

Source context: [OWASP-LLM2026](#)

AGT-04

HYBRID VALIDATION

Protect retrieval and persistent memory

- ☐ Enforce document and memory ACLs at retrieval and update time, including vector search metadata filters.
- ☐ Test poisoned memory persistence, cross-tenant retrieval, provenance loss, deletion, and stale privileged context.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [OWASP-AGENTIC](#) / [CSA-MAESTRO](#) / [CIS-AGENTS-2026](#)

05 / CONTROL CATALOG

Agent behavior and context

AGT-05 - AGT-06 / Check applicability, retain evidence, and resolve exceptions.

AGT-05

DYNAMIC VALIDATION

Keep objectives and authority bounded across agents

- ☐ Constrain delegated tasks and verify messages against authenticated senders, expected schemas, and allowed transitions.
- ☐ Test impersonation, conflicting instructions, cascading failure, and privilege growth across handoffs.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [OWASP-AGENTIC](#) / [CSA-MAESTRO](#) / [OWASP-AISVS-C9](#)

AGT-06

HYBRID VALIDATION

Protect agent configuration and skills

- ☐ Inventory skill files, prompts, hooks, memory seed files, MCP configuration, and other executable workflow inputs.
- ☐ Require review for changes that add commands, access, or persistence; external repository text cannot become trusted policy.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI043, AI044, AI045 (partial)

Source context: [MITRE-ATLAS](#) / [NCSC-SECURE-AI](#) / [AGENT-SKILLS-SPEC](#) / [AIUC1-Q3-2026](#)

05 / CONTROL CATALOG

Agent behavior and context

AGT-07 - AGT-07 / Check applicability, retain evidence, and resolve exceptions.

AGT-07

DYNAMIC VALIDATION

Prevent sensitive context leaving through legitimate tools

- ☐ Apply destination and data policies to URLs, searches, tickets, messages, uploads, and telemetry generated by agents.
- ☐ Use canary data to test encoded leakage and combinations of otherwise permitted tools.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.**Static coverage:** AI044 (partial)**Source context:** [OWASP-MCP-CS](#)

ASSESSMENT RECORD

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Owner / review date

Evidence / test run

Result / exception / next action

05 / CONTROL CATALOG

Execution and application security

EXEC-01 - EXEC-02 / Check applicability, retain evidence, and resolve exceptions.

EXEC-01

HYBRID VALIDATION

Prevent shell and command injection

- ☐ Find shell execution and constructed command strings; use fixed executables, argument arrays, and allowed argument values.
- ☐ Test untrusted tool inputs containing shell syntax, option injection, command substitution, and hostile filenames.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI002, AI003, AI012 (partial)

Source context: [OWASP-MCP10](#) / [OWASP-OUTPUT2025](#)

EXEC-02

HYBRID VALIDATION

Constrain generated-code execution

- ☐ Locate eval, exec, dynamic imports, templates, notebooks, and interpreter tools accepting model or user content.
- ☐ Run required code execution in a disposable restricted environment with explicit filesystem, network, CPU, and time limits.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI001, AI013 (partial)

Source context: [OWASP-AGENT-CS](#) / [OWASP-AISVS-C9](#) / [AIUC1-Q3-2026](#)

05 / CONTROL CATALOG

Execution and application security

EXEC-03 - EXEC-04 / Check applicability, retain evidence, and resolve exceptions.

EXEC-03

HYBRID VALIDATION

Parameterize database and query operations

- ☐ Use bound query parameters and allowed query shapes; inspect SQL, NoSQL, graph, and search-language construction.
- ☐ Separate read/write database identities and test whether generated queries can escape permitted objects or operations.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI036 (partial)

Source context: [OWASP-OUTPUT2025](#)

EXEC-04

HYBRID VALIDATION

Constrain file and archive access

- ☐ Resolve and enforce allowed paths at access time; constrain uploads, downloads, extraction, temporary files, and permissions.
- ☐ Test symlink races, archive traversal, absolute paths, overwrite attempts, and secret-directory reads.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI015, AI016, AI037, AI047 (partial)

Source context: [MCP-ROOTS](#) / [JOINT-DEPLOY](#)

05 / CONTROL CATALOG

Execution and application security

EXEC-05 - EXEC-06 / Check applicability, retain evidence, and resolve exceptions.

EXEC-05

HYBRID VALIDATION

Prevent SSRF and unsafe network destinations

- ☐ Restrict destinations and schemes at connection time; revalidate DNS resolution and each redirect.
- ☐ Test loopback, private/link-local IPv4 and IPv6, cloud metadata, alternate encodings, and redirect-to-private cases.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.**Static coverage:** AI014 (partial)**Source context:** [MCP-SECURITY](#)**EXEC-06**

HYBRID VALIDATION

Reject unsafe parsing and deserialization

- ☐ Inspect pickle, unsafe YAML, object deserialization, XML entity expansion, and unconstrained recursive parsers.
- ☐ Use data-only formats with byte, nesting, and type limits; test malformed input and expansion attacks.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.**Static coverage:** AI004, AI005, AI035 (partial)**Source context:** [NIST-SSDF](#)

05 / CONTROL CATALOG

Execution and application security

EXEC-07 - EXEC-07 / Check applicability, retain evidence, and resolve exceptions.

EXEC-07

HYBRID VALIDATION

Render model and tool output safely

- ☐ Use context-specific escaping for HTML/Markdown, avoid unsafe DOM sinks, and validate links and embedded media.
- ☐ Test active SVG/HTML, malicious URLs, terminal escapes, and spreadsheet formulas in exported reports.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.**Static coverage:** AI039, AI040 (partial)**Source context:** [OWASP-OUTPUT2025](#)

ASSESSMENT RECORD

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Owner / review date

Evidence / test run

Result / exception / next action

05 / CONTROL CATALOG

Data and privacy

DATA-01 - DATA-02 / Check applicability, retain evidence, and resolve exceptions.

DATA-01

STATIC VALIDATION

Detect and remove embedded credentials

- ☐ Inspect code, examples, configuration, notebooks, test fixtures, and generated artifacts for secret-like values.
- ☐ Verify actual exposures with the owner, rotate real credentials, and remove them from reachable history and artifacts.

Evidence: Code/configuration location, applicable version, and reviewer disposition.

Static coverage: AI010, AI011, AI030, AI034 (partial)

Source context: [JOINT-DEPLOY](#) / [AIUC1-Q3-2026](#)

DATA-02

MANUAL VALIDATION

Minimize data sent to models and gateways

- ☐ Map which prompts, tool outputs, memory, and code leave the environment and identify the receiving provider/gateway.
- ☐ Document allowed data classes, processing location, retention, and training use; redact before transmission where required.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-GENAI](#) / [CSA-AICM](#) / [CIS-AGENTS-2026](#) / [AIUC1-Q3-2026](#)

05 / CONTROL CATALOG

Data and privacy

DATA-03 - DATA-04 / Check applicability, retain evidence, and resolve exceptions.

DATA-03

HYBRID VALIDATION

Protect logs, traces, and error responses

- ☐ Redact secrets and sensitive payloads before logs, traces, exception strings, and dashboards persist them.
- ☐ Test the failure paths and provider errors as well as success paths; restrict access and export destinations.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI009, AI033 (partial)

Source context: [OWASP-MCP10](#) / [CIS-AGENTS-2026](#) / [CIS-MCP-2026](#) / [AIUC1-Q3-2026](#)

DATA-04

MANUAL VALIDATION

Track provenance and integrity of AI data

- ☐ Record source, owner, version, transformation history, and integrity evidence for datasets, retrieval corpora, and memory seeds.
- ☐ Quarantine unexpected changes and test how poisoned or stale data is detected, removed, and replaced.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NSA-DATA](#) / [CSA-AICM](#) / [NIST-SSDF-AI](#)

05 / CONTROL CATALOG

Data and privacy

DATA-05 - DATA-06 / Check applicability, retain evidence, and resolve exceptions.

DATA-05

DYNAMIC VALIDATION

Enforce retention and deletion across copies

- ☐ Apply expiry and deletion to prompts, embeddings, caches, memory, tool artifacts, backups, and provider-held data.
- ☐ Verify deleting a source record removes or invalidates dependent retrieval content and access grants.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-GENAI](#) / [CIS-AGENTS-2026](#)

DATA-06

HYBRID VALIDATION

Protect stored data and keys

- ☐ Check transport verification, storage access controls, encryption settings, key separation, and backup permissions.
- ☐ Validate key rotation and denied access using a principal outside the authorized tenant or operational role.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NSA-DATA](#) / [JOINT-DEPLOY](#) / [CSA-CCM](#) / [CIS-AGENTS-2026](#)

05 / CONTROL CATALOG

Supply chain

SUP-01 - SUP-02 / Check applicability, retain evidence, and resolve exceptions.

SUP-01

STATIC VALIDATION

Pin and inventory executable dependencies

- ☐ Review lockfiles and exact versions or immutable digests for packages, images, MCP servers, models, and plugins.
- ☐ Flag runtime installs, floating tags, remote scripts, and dependency sources outside approved registries.

Evidence: Code/configuration location, applicable version, and reviewer disposition.

Static coverage: AI018, AI024, AI025 (partial)

Source context: [NIST-SSDF](#) / [NIST-SSDF-AI](#) / [AIUC1-Q3-2026](#)

SUP-02

MANUAL VALIDATION

Check vulnerability and maintenance exposure

- ☐ Run appropriate package/container advisory tools against resolved dependencies and save database date and tool version.
- ☐ Triage reachability, fix availability, support status, and transitive dependencies; source pattern scans do not establish CVE coverage.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NCSC-SECURE-AI](#) / [CSA-AISMM](#) / [CIS-MCP-2026](#) / [OPENSSF-BASELINE-202608](#)

05 / CONTROL CATALOG

Supply chain

SUP-03 - SUP-04 / Check applicability, retain evidence, and resolve exceptions.

SUP-03

MANUAL VALIDATION

Verify artifact identity and provenance

- ☐ Verify publisher identity, hashes/signatures, build provenance, and intended origin before enabling artifacts.
- ☐ Review model loading and serialization behavior; an integrity hash cannot make an untrusted publisher safe.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: AI019, AI024, AI035 (partial)

Source context: NSA-DATA / NIST-SSDF / CSA-AICM / CSA-MAESTRO / NIST-SSDF-AI / OWASP-AISVS-C10 / SLSA-12 / OPENSSEF-MODEL-SIGNING

SUP-04

HYBRID VALIDATION

Protect build, release, and configuration changes

- ☐ Restrict CI credentials and workflow permissions; review actions, build scripts, and release provenance.
- ☐ Prevent untrusted contributions from executing with deployment secrets or changing approved agent policies.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI020 (partial)

Source context: NIST-SSDF / NIST-SSDF-AI / CIS-MCP-2026 / OPENSSEF-BASELINE-202608 / SLSA-12

05 / CONTROL CATALOG

Supply chain

SUP-05 - SUP-06 / Check applicability, retain evidence, and resolve exceptions.

SUP-05

HYBRID VALIDATION

Harden runtime isolation and deployment defaults

- ☐ Review root/privileged containers, host mounts, Docker sockets, unrestricted egress, debug mode, and public management endpoints.
- ☐ Separate agent execution from control-plane credentials and audit storage; test isolation in the deployed environment.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: AI021, AI022, AI023, AI031, AI042 (partial)

Source context: [JOINT-DEPLOY](#) / [CSA-CCM](#) / [CSA-MAESTRO](#)

SUP-06

DYNAMIC VALIDATION

Separate and protect model development environments

- ☐ When training or fine-tuning, isolate development, evaluation, and production identities, data access, registries, and release permissions.
- ☐ Protect datasets, weights, adapters, and configuration separately; monitor modifications and demonstrate that an untrusted training job cannot replace an approved production artifact.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-SSDF-AI](#)

05 / CONTROL CATALOG

Operations and resilience

OPS-01 - OPS-02 / Check applicability, retain evidence, and resolve exceptions.

OPS-01

HYBRID VALIDATION

Produce attributable audit events

- ☐ Record initiating identity, delegated identity, tool/server/version, approved arguments, decision, outcome, and correlation identifiers.
- ☐ Protect log integrity and clock consistency; ensure agents cannot erase their own action history.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [OWASP-MCP10](#) / [CSA-CCM](#) / [CSA-AGENT-IAM](#) / [CIS-AGENTS-2026](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#)

OPS-02

HYBRID VALIDATION

Bound work, spending, and concurrency

- ☐ Set server-enforced limits for iterations, tokens, tool calls, recursion, parallelism, bytes, cost, and elapsed time.
- ☐ Exercise stuck loops and amplification paths; verify limits apply across retries and child agents.

Evidence: Source/configuration trace plus negative-test result from the effective deployment.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [ASD-HARNESS](#) / [CSA-SCOPING](#) / [OWASP-AISVS-C9](#) / [OWASP-AISVS-C10](#)

05 / CONTROL CATALOG

Operations and resilience

OPS-03 - OPS-04 / Check applicability, retain evidence, and resolve exceptions.

OPS-03

DYNAMIC VALIDATION

Make cancellation and shutdown effective

- ☐ Provide a tested stop mechanism that revokes work, credentials, queued actions, and child tasks.
- ☐ Measure stop latency and verify cancelled or disconnected requests cannot later commit prohibited side effects.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [JOINT-AGENTIC](#) / [CSA-AGENT-IAM](#) / [OWASP-AISVS-C9](#)

OPS-04

DYNAMIC VALIDATION

Fail safely and prevent duplicate side effects

- ☐ Deny or escalate when policy, identity, or approval checks fail; prevent fallback paths from widening privilege.
- ☐ Test outages, partial failures, timeouts, retries, idempotency, and recovery of transactions with external effects.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [ASD-HARNESS](#)

05 / CONTROL CATALOG

Operations and resilience

OPS-05 - OPS-06 / Check applicability, retain evidence, and resolve exceptions.

OPS-05**DYNAMIC VALIDATION**

Monitor behavior and support rollback

- ☐ Alert on unusual tool use, new destinations, scope growth, repeated denials, cost spikes, and unexpected state changes.
- ☐ Validate alerts with seeded events and test rollback of models, prompts, tools, policies, and poisoned memory.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NCSC-SECURE-AI](#) / [CSA-CCM](#) / [CSA-AISMM](#)

OPS-06**MANUAL VALIDATION**

Practice incident response and disclosure

- ☐ Maintain procedures to isolate agents, revoke credentials, preserve evidence, notify owners, and recover trusted state.
- ☐ Exercise an AI-specific incident and define approved vulnerability/intelligence-sharing channels without exposing sensitive evidence.

Evidence: Named owner, approved scope, evidence record, review date, and exceptions.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [CISA-JCDC](#) / [CSA-AICM](#) / [CSA-CCM](#) / [CSA-AISMM](#) / [OPENSSF-BASELINE-202608](#)

05 / CONTROL CATALOG

Security validation

TEST-01 - TEST-02 / Check applicability, retain evidence, and resolve exceptions.

TEST-01

DYNAMIC VALIDATION

Measure prompt-injection security and useful task completion

- ☐ Run paired benign and adversarial workflows with representative tools and attacker-controlled external content.
- ☐ Record attacker success, authorized task success, blocked benign actions, and the observed unauthorized side effect.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [BENCH-AGENTDOJO](#) / [BENCH-INJECAGENT](#) / [MITRE-ATLAS-202609](#)

TEST-02

DYNAMIC VALIDATION

Test adaptively and repeat scenarios

- ☐ Vary payload wording, placement, modality, attacker knowledge, and attempts; rerun after changing defenses.
- ☐ Report per-scenario outcomes, sample counts, attack budget, uncertainty, and model/harness versions instead of only an average.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-HIJACK-EVAL](#) / [MITRE-ATLAS-202609](#)

05 / CONTROL CATALOG

Security validation

TEST-03 - TEST-04 / Check applicability, retain evidence, and resolve exceptions.

TEST-03

DYNAMIC VALIDATION

Exercise MCP authentication and protocol abuse

- ☐ In an isolated test environment, exercise invalid tokens, wrong audiences, origin abuse, malicious servers, and protocol fuzzing.
- ☐ Adapt scenarios to the deployed MCP revision and transports; preserve request/response evidence with secrets removed.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [BENCH-MCPSECBENCH](#) / [MCP-AUTH](#) / [CIS-MCP-2026](#)

TEST-04

DYNAMIC VALIDATION

Verify cross-tenant and cross-agent isolation

- ☐ Create two principals and attempt cross-access to objects, memory, caches, handles, subscriptions, and execution results.
- ☐ Repeat after reconnect, delegation, failed authentication, concurrent requests, and privilege revocation.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [BENCH-ASB](#)

05 / CONTROL CATALOG

Security validation

TEST-05 - TEST-06 / Check applicability, retain evidence, and resolve exceptions.

TEST-05

DYNAMIC VALIDATION

Test approval, policy, and sandbox bypass

- ☐ Use explicit forbidden-action canaries to verify enforcement survives hostile content, tool substitution, and delayed execution.
- ☐ Confirm paths through retries, fallback models, alternate tools, and child agents enforce the same boundary.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [OWASP-AGENT-CS](#) / [CIS-MCP-2026](#)

TEST-06

DYNAMIC VALIDATION

Verify conventional application security

- ☐ Run language-aware SAST, dependency review, secret scanning, and authorized integration tests for exposed services.
- ☐ Exercise reachable injection, XSS, SSRF, path traversal, deserialization, and access-control risks with safe fixtures.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-SSDF](#) / [OWASP-OUTPUT2025](#) / [NIST-SSDF-AI](#) / [CIS-MCP-2026](#) / [AIUC1-Q3-2026](#)

05 / CONTROL CATALOG

Security validation

TEST-07 - TEST-08 / Check applicability, retain evidence, and resolve exceptions.

TEST-07

DYNAMIC VALIDATION

Validate resource exhaustion and observability

- ❑ Simulate oversized messages, slow peers, streaming floods, repeated errors, runaway agents, and unavailable dependencies.
- ❑ Verify quotas, shutdown, telemetry, and alerts work together without leaking payloads or losing attribution.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [ASD-HARNESS](#)

TEST-08

DYNAMIC VALIDATION

Evaluate the optional security judge itself

- ❑ Treat repository text and model verdicts as untrusted; test prompt injection, fabricated locations, invalid JSON, and timeouts.
- ❑ Compare against labeled cases, retain deterministic results, document model variability, and never equate a judge approval with control validation.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [NIST-HIJACK-EVAL](#) / [NIST-GENAI](#)

05 / CONTROL CATALOG

Security validation

TEST-09 - TEST-09 / Check applicability, retain evidence, and resolve exceptions.

TEST-09

DYNAMIC VALIDATION

Evaluate model-layer attacks for hosted or customized models

- ☐ Where applicable, assess suspicious triggers, model extraction, and disclosure of private training examples under an explicit attacker access and query budget.
- ☐ Retest acquired or retrained models and adapters before release; document measured failures, model identity, coverage limits, and accepted residual risk.

Evidence: Pinned scenario, expected policy, observed effects, and repeatable test results.

Static coverage: No mapped rule; obtain review or runtime evidence.

Source context: [CSA-MAESTRO](#) / [NIST-SSDF-AI](#)

ASSESSMENT RECORD

Use this space to record evidence for the control above. A source citation or a clean scan alone is not a validation result.

Owner / review date

Evidence / test run

Result / exception / next action

06 / AUTOMATION INDEX

What the scanner can flag

All rules are static signals. Confidence and context determine the next investigation step; no rule proves full control effectiveness.

RULE	SEVERITY	RISK PATTERN
AI001	HIGH	Dynamic Python code execution
AI002	HIGH	Dynamic command executed through a shell
AI003	HIGH	Dynamic os shell command
AI004	HIGH	Unsafe YAML object construction
AI005	HIGH	Executable deserialization requires trusted inputs
AI006	HIGH	TLS certificate verification disabled
AI007	MEDIUM	Wildcard cross-origin access
AI008	LOW	Service binds to all network interfaces
AI009	MEDIUM	Application debug mode enabled
AI010	HIGH	Credential-like literal in source or configuration
AI011	CRITICAL	Private key material in repository
AI012	HIGH	Dynamic JavaScript shell execution
AI013	HIGH	Dynamic JavaScript code execution
AI014	MEDIUM	Externally influenced outbound request

The machine-readable catalog maps these rules to selected controls. A rule match requires investigation. A suppressed finding remains visible as an accepted exception; an unmapped or untested control remains unvalidated.

06 / AUTOMATION INDEX

What the scanner can flag

All rules are static signals. Confidence and context determine the next investigation step; no rule proves full control effectiveness.

RULE	SEVERITY	RISK PATTERN
AI015	MEDIUM	Externally influenced filesystem path
AI016	MEDIUM	Race-prone temporary filename creation
AI017	HIGH	JWT verification explicitly bypassed
AI018	MEDIUM	MCP package runner resolves an unpinned artifact
AI019	HIGH	Remote download executed directly by a shell
AI020	MEDIUM	GitHub Action reference is not pinned to a commit
AI021	MEDIUM	Container explicitly runs as root
AI022	HIGH	Privileged container or privilege escalation enabled
AI023	HIGH	Container runtime socket exposed
AI024	LOW	Container image is not digest pinned
AI025	LOW	Direct dependency is not exactly pinned
AI026	HIGH	Authentication explicitly disabled
AI027	HIGH	Wildcard tool or permission authorization
AI028	HIGH	MCP token passthrough enabled

The machine-readable catalog maps these rules to selected controls. A rule match requires investigation. A suppressed finding remains visible as an accepted exception; an unmapped or untested control remains unvalidated.

06 / AUTOMATION INDEX

What the scanner can flag

All rules are static signals. Confidence and context determine the next investigation step; no rule proves full control effectiveness.

RULE	SEVERITY	RISK PATTERN
AI029	HIGH	Remote MCP URL uses plaintext HTTP
AI030	MEDIUM	Browser-side API credential access explicitly enabled
AI031	HIGH	Agent approval or sandbox safeguard explicitly bypassed
AI032	MEDIUM	External data inserted into privileged LLM instructions
AI033	MEDIUM	Potential secret included in logging call
AI034	HIGH	Credential-like value embedded in URL query
AI035	HIGH	Unsafe model checkpoint deserialization
AI036	HIGH	SQL query constructed with interpolation
AI037	MEDIUM	Archive extraction without an explicit safe filter
AI038	MEDIUM	Cryptographically weak randomness used for a security value
AI039	HIGH	Server-Side Template rendered from a dynamic template
AI040	MEDIUM	Unsafe HTML rendering boundary
AI041	HIGH	MCP inspector authorization explicitly disabled
AI042	HIGH	Container uses host namespaces

The machine-readable catalog maps these rules to selected controls. A rule match requires investigation. A suppressed finding remains visible as an accepted exception; an unmapped or untested control remains unvalidated.

06 / AUTOMATION INDEX

What the scanner can flag

All rules are static signals. Confidence and context determine the next investigation step; no rule proves full control effectiveness.

RULE	SEVERITY	RISK PATTERN
AI043	HIGH	Instruction-hierarchy override in agent-facing text
AI044	HIGH	Sensitive-data transfer instruction
AI045	HIGH	Covert action or approval-bypass instruction
AI046	MEDIUM	Read-only tool annotation conflicts with declared or observed writes
AI047	MEDIUM	Filesystem permissions explicitly allow writes by everyone

The machine-readable catalog maps these rules to selected controls. A rule match requires investigation. A suppressed finding remains visible as an accepted exception; an unmapped or untested control remains unvalidated.

06 / EXACT SCAN CONTRACT / AI001

Dynamic Python code execution

AI001 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Nonliteral eval/exec calls, including recognized builtins aliases and constant getattr names.

ALGORITHM + SUPPORTED PATHS

AST call resolution and bounded local constant/alias tracking. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent can convert a prompt or MCP tool result into an eval/exec argument. If that value reaches this call, model influence becomes Python execution with the tool process identity.

CAN MISS OR MISCLASSIFY

Dynamic imports, computed attributes and cross-function input flow are not completely resolved; a dynamic argument alone does not prove attacker control.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Replace dynamic execution with an allowlisted operation dispatcher; use ast.literal_eval only for bounded literal parsing. Isolate unavoidable execution with no ambient credentials and strict resource limits.

Partial control mappings: EXEC-02. A mapping does not satisfy the full control.

Primary control source: OWASP-AGENT-CS

Thematic alignment: OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: TECH-PYTHON-SECURITY / OWASP-LLM2026

Explore: invscan --explain-scan AI001

Run: invscan TARGET --scans AI001

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI002

Dynamic command executed through a shell

AI002 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Dynamic subprocess shell=True, asyncio.create_subprocess_shell, or an explicit recognized shell executable with its command option.

ALGORITHM + SUPPORTED PATHS

AST call/argument matching and bounded local tracking. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent or MCP tool often translates model-selected operations into subprocess commands. Untrusted arguments embedded in a shell string can turn one permitted tool action into additional commands.

CAN MISS OR MISCLASSIFY

Does not establish command reachability, input sanitization across functions, executable trust, or sandbox effectiveness.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Pass an argument list with shell=False, allowlist executable names and options, and enforce working-directory and resource restrictions.

Partial control mappings: EXEC-01. A mapping does not satisfy the full control.

Primary control source: OWASP-MCP10 / OWASP-OUTPUT2025

Rule technical reference: TECH-PYTHON-SECURITY / OWASP-LLM2026

Explore: invscan --explain-scan AI002

Run: invscan TARGET --scans AI002

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI003

Dynamic os shell command

AI003 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Dynamic os.system/os.popen arguments through recognized imports and aliases.

ALGORITHM + SUPPORTED PATHS

AST resolved-API matching and constant exclusion. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An MCP command tool may interpolate planner output into os.system or os.popen. Those APIs introduce a shell boundary that can expand attacker-controlled syntax into the server account privileges.

CAN MISS OR MISCLASSIFY

A dangerous shell boundary is identified, not complete upstream taint or production exploitability.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use subprocess with a fixed executable and argument list, shell=False, and explicit allowed options.

Partial control mappings: EXEC-01. A mapping does not satisfy the full control.

Primary control source: [OWASP-MCP10](#) / [OWASP-OUTPUT2025](#)

Rule technical reference: [TECH-PYTHON-SECURITY](#) / [OWASP-LLM2026](#)

Explore: invscan --explain-scan AI003

Run: invscan TARGET --scans AI003

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI004

Unsafe YAML object construction

AI004 / HIGH / deserialization

WHAT THE DETERMINISTIC RULE DETECTS

Recognized YAML load/load_all/unsafe_load calls without an explicitly recognized SafeLoader, CSafeLoader, BaseLoader or CBaseLoader.

ALGORITHM + SUPPORTED PATHS

AST loader allowlist. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agents and MCP servers often read tool manifests, workflows or uploaded configuration as YAML. An executable loader can convert attacker-influenced configuration into Python objects before tool authorization runs.

CAN MISS OR MISCLASSIFY

User-defined wrapper safety and every third-party YAML API are outside the allowlist; the actual input's trust is not established.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use yaml.safe_load or SafeLoader/CSafeLoader and validate the parsed schema and size.

Partial control mappings: EXEC-06. A mapping does not satisfy the full control.

Primary control source: NIST-SSDF

Rule technical reference: TECH-PYYAML

Explore: invscan --explain-scan AI004

Run: invscan TARGET --scans AI004

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI005

Executable deserialization requires trusted inputs

AI005 / HIGH / deserialization

WHAT THE DETERMINISTIC RULE DETECTS

pickle/dill/cloudpickle load/loads and joblib.load calls.

ALGORITHM + SUPPORTED PATHS

AST executable-deserializer API matching. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A cached agent state, plugin artifact or MCP-uploaded file may reach pickle-compatible loading. Deserialization can execute producer-selected code before downstream data validation.

CAN MISS OR MISCLASSIFY

Authenticated trusted artifacts may be intentional; this does not inspect serialized objects or prove attacker control.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use a nonexecutable serialization format with schema validation. If compatibility requires pickle, accept only authenticated artifacts from a strictly controlled producer.

Partial control mappings: EXEC-06. A mapping does not satisfy the full control.

Primary control source: [NIST-SSDF](#)

Rule technical reference: [TECH-PYTHON-SECURITY](#)

Explore: invscan --explain-scan AI005

Run: invscan TARGET --scans AI005

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI006

TLS certificate verification disabled

AI006 / HIGH / transport security

WHAT THE DETERMINISTIC RULE DETECTS

Explicit TLS verification bypass in supported Python HTTP clients/SSL, JavaScript rejectUnauthorized, NODE_TLS_REJECT_UNAUTHORIZED and supported configuration fields.

ALGORITHM + SUPPORTED PATHS

Typed literal and recognized network/configuration API checks. Profiles: python_ast, javascript_lexical, json_structured, configuration_lexical. Image evidence contexts: final_filesystem, runtime_configuration.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

The affected connection can carry model credentials, prompts, MCP tool requests or retrieved evidence. Disabling TLS verification lets a reachable intermediary impersonate that peer and influence both data and agent decisions.

CAN MISS OR MISCLASSIFY

Does not establish effective CA trust, hostname validation, gateway behavior or an override supplied at deployment.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Enable certificate verification and install the correct CA bundle; use a trusted private CA for custom gateways.

Partial control mappings: MCP-01. A mapping does not satisfy the full control.

Primary control source: MCP-HTTP

Thematic alignment: OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY / CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI006

Run: invscan TARGET --scans AI006

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI007

Wildcard cross-origin access

AI007 / MEDIUM / network exposure

WHAT THE DETERMINISTIC RULE DETECTS

Wildcard origins in supported Python CORS calls, JavaScript fields and structured configuration.

ALGORITHM + SUPPORTED PATHS

Literal origin-list matching. Profiles: python_ast, javascript_lexical, json_structured. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A browser can reach some local or remote MCP HTTP services. A broad cross-origin policy may expose tool responses or weaken the browser-facing boundary around operations running with the user credentials.

CAN MISS OR MISCLASSIFY

Does not infer authentication, browser reachability, credential mode or framework-specific CORS response behavior.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Allow only required origins, validate Origin on HTTP MCP endpoints, and combine origin checks with authentication. Do not combine wildcard origins with credentials.

Partial control mappings: MCP-01. A mapping does not satisfy the full control.

Primary control source: MCP-HTTP

Thematic alignment: OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY

Explore: invscan --explain-scan AI007

Run: invscan TARGET --scans AI007

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI008

Service binds to all network interfaces

AI008 / LOW / network exposure

WHAT THE DETERMINISTIC RULE DETECTS

Wildcard host bindings such as 0.0.0.0 or :: in supported calls and configuration keys.

ALGORITHM + SUPPORTED PATHS

Literal listener/bind matching. Profiles: python_ast, javascript_lexical, json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A broadly bound agent dashboard or MCP HTTP server may expose its tool authority to additional networks. The bind literal does not establish whether firewalls, published ports or authentication actually permit access.

CAN MISS OR MISCLASSIFY

Binding is an exposure review signal; ingress restrictions, authentication and actual remote reachability require deployment evidence.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Bind local-only tools to loopback. For intentional remote services, verify ingress policy, TLS, authentication, and origin validation.

Partial control mappings: MCP-01. A mapping does not satisfy the full control.

Primary control source: MCP-HTTP

Thematic alignment: OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY / CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI008

Run: invscan TARGET --scans AI008

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI009

Application debug mode enabled

AI009 / MEDIUM / configuration

WHAT THE DETERMINISTIC RULE DETECTS

Recognized debug=True calls/assignments and supported debug/FLASK_DEBUG configuration values.

ALGORITHM + SUPPORTED PATHS

Explicit debug-setting matching. Profiles: python_ast, json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent API or MCP server may handle secrets, prompts and privileged tools. A reachable development debugger or verbose exception page can reveal that state and, in some frameworks, permit code execution.

CAN MISS OR MISCLASSIFY

A setting may be development-only; environment, framework defaults and production overrides are not proven.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Disable debug mode in deployed services and return sanitized errors while retaining protected server-side diagnostics.

Partial control mappings: DATA-03. A mapping does not satisfy the full control.

Primary control source: OWASP-MCP10

Thematic alignment: CIS-AGENTS-2026 / CIS-MCP-2026 / AIUC1-Q3-2026

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI009

Run: invscan TARGET --scans AI009

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI010

Credential-like literal in source or configuration

AI010 / HIGH / secrets

WHAT THE DETERMINISTIC RULE DETECTS

Nonplaceholder credential-named literals, supported token formats and literal bearer credentials; explicit environment-variable-name fields are excluded.

ALGORITHM + SUPPORTED PATHS

Credential-name/value heuristics and bounded token patterns. Profiles: `python_ast`, `json_structured`, `generic_text`. Image evidence contexts: `final_filesystem`, `runtime_configuration`, `retained_layer`, `build_history`.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A hardcoded model, gateway or downstream-tool key can grant access outside the intended agent session and tenant. Repository or image readers may obtain the credential independently of the agent approval path.

CAN MISS OR MISCLASSIFY

May miss unusual secret formats and may flag realistic test data; never validates credentials remotely or proves they are live.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

If real, revoke and rotate the credential, remove it from source and history, and load it from a secret manager or environment variable with least privilege.

Partial control mappings: AUTH-05, DATA-01. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH-SEC / JOINT-DEPLOY

Thematic alignment: CSA-AGENT-IAM / CIS-AGENTS-2026 / CIS-MCP-2026 / OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: CISA-SECURE-BY-DESIGN / MCP-SECURITY

Explore: `invscan --explain-scan AI010`

Run: `invscan TARGET --scans AI010`

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI011

Private key material in repository

AI011 / CRITICAL / secrets

WHAT THE DETERMINISTIC RULE DETECTS

Private-key PEM headers accompanied by plausible encoded key data.

ALGORITHM + SUPPORTED PATHS

PEM marker plus plausible encoded-material check. Profiles: generic_text. Image evidence contexts: final_filesystem, runtime_configuration, retained_layer, build_history.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A signing or TLS key used by an agent gateway or MCP identity can let another party impersonate a service or forge trusted artifacts. A repository/image copy can outlive the file currently deployed.

CAN MISS OR MISCLASSIFY

Does not establish cryptographic validity, key use or test-versus-production ownership.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Determine whether the key was used, revoke or rotate it if necessary, remove it from version history, and use a managed key store.

Partial control mappings: AUTH-05, DATA-01. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH-SEC / JOINT-DEPLOY

Thematic alignment: CSA-AGENT-IAM / CIS-AGENTS-2026 / CIS-MCP-2026 / OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI011

Run: invscan TARGET --scans AI011

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI012

Dynamic JavaScript shell execution

AI012 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Dynamic Node child-process shell execution APIs in supported import and call forms.

ALGORITHM + SUPPORTED PATHS

Tokenized child-process imports/calls with bounded alias handling. Profiles: javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent or MCP tool can pass model-selected text to child_process.exec or a shell-enabled spawn. Shell parsing can grant extra command execution under the server account.

CAN MISS OR MISCLASSIFY

Not a full JavaScript/TypeScript parser or interprocedural taint engine; computed wrappers and dynamic imports may be missed.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use execFile or spawn with a fixed executable, validated argument array, and shell disabled; constrain tools by policy before execution.

Partial control mappings: EXEC-01. A mapping does not satisfy the full control.

Primary control source: OWASP-MCP10 / OWASP-OUTPUT2025

Rule technical reference: TECH-NODE-PROCESS / OWASP-LLM2026

Explore: invscan --explain-scan AI012

Run: invscan TARGET --scans AI012

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI013

Dynamic JavaScript code execution

AI013 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Dynamic JavaScript eval or Function constructor arguments.

ALGORITHM + SUPPORTED PATHS

Balanced token call matching and literal exclusion. Profiles: javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

If planner output or tool results reach eval or a Function constructor, agent-controlled text can execute with the host application privileges rather than merely describe an operation.

CAN MISS OR MISCLASSIFY

Wrapper calls, runtime bindings, reachability and input provenance may remain unknown.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use validated structured data and allowlisted operations instead of evaluating generated code.

Partial control mappings: EXEC-02. A mapping does not satisfy the full control.

Primary control source: OWASP-AGENT-CS

Thematic alignment: OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: OWASP-LLM2026

Explore: invscan --explain-scan AI013

Run: invscan TARGET --scans AI013

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI014

Externally influenced outbound request

AI014 / MEDIUM / ssrf

WHAT THE DETERMINISTIC RULE DETECTS

Recognized HTTP sinks with external influence, including supported Python FastMCP/LangChain tool parameters, conventional unbound mcp/server.tool registration forms, local function aliases and argument/return flow. Literal JavaScript server/mcp.registerTool callbacks seed simple inputs and direct-return named/arrow wrappers propagate to fetch/axios. Exact Python string equality or literal tuple/set guards constrain destination selection.

ALGORITHM + SUPPORTED PATHS

Registered input boundaries, call-site propagation and exact literal constraints. Profiles: python_ast, javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A fetch/search/browser MCP tool can turn a model-selected URL into a request from a privileged network location. An attacker may use retrieved instructions or tool arguments to target internal services or metadata.

CAN MISS OR MISCLASSIFY

Python calls are bounded to depth 8 and 512 expansions; splats, recursion, indirect coroutine/generator execution and global/nonlocal side effects leave gaps while retaining findings. Unknown decorators, methods, cross-file flow and dynamic dispatch remain unresolved. JS requires narrow literal registration and direct-return wrappers. Conventional receiver names indicate a declared boundary, not verified SDK identity. Fixed destinations do not prove DNS/redirect safety.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Allowlist destinations and schemes, resolve and validate addresses on each connection and redirect, block metadata/private destinations unless explicitly needed, and enforce egress policy.

Partial control mappings: EXEC-05. A mapping does not satisfy the full control.

Primary control source: MCP-SECURITY

Rule technical reference: MCP-SECURITY

Explore: invscan --explain-scan AI014

Run: invscan TARGET --scans AI014

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI015

Externally influenced filesystem path

AI015 / MEDIUM / filesystem

WHAT THE DETERMINISTIC RULE DETECTS

Supported filesystem operations receiving external data through Python tool parameters/local calls or literal JavaScript registerTool callbacks/direct-return wrappers with recognized filesystem imports. Python literal finite string maps can constrain a selected path; aliases, mutations and unknown-call escapes invalidate that local proof.

ALGORITHM + SUPPORTED PATHS

Registered input boundaries and bounded filesystem argument propagation. Profiles: python_ast, javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent and MCP file tools can translate model-selected names into reads, writes or deletions. Without confinement, a prompt or tool result may reach server credentials, another tenant workspace or executable files.

CAN MISS OR MISCLASSIFY

Canonicalization, symlink races, unknown decorators/methods, cross-file validation and effective confinement need review. A finite map does not prove tenant authorization. Named callback references, complex JS callbacks/wrappers and unmodeled dispatch remain outside this bounded analysis; unsupported call binding and exhausted budgets leave gaps.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Resolve paths under a fixed workspace root, reject paths outside it after canonicalization, handle symlinks safely, and give the process only required filesystem permissions.

Partial control mappings: MCP-08, EXEC-04. A mapping does not satisfy the full control.

Primary control source: MCP-ROOTS / JOINT-DEPLOY

Rule technical reference: MCP-SECURITY / OWASP-LLM2026

Explore: invscan --explain-scan AI015

Run: invscan TARGET --scans AI015

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI016

Race-prone temporary filename creation

AI016 / MEDIUM / filesystem

WHAT THE DETERMINISTIC RULE DETECTS

tempfile.mktemp usage, which returns a name without atomically creating the file.

ALGORITHM + SUPPORTED PATHS

Resolved AST API matching. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent or MCP worker may stage sensitive tool inputs or outputs in temporary files. A name-only reservation lets another local process substitute a file or symlink before the worker opens it.

CAN MISS OR MISCLASSIFY

Does not prove a concurrent attacker, downstream file permissions or whether the path is subsequently used.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use TemporaryFile, NamedTemporaryFile, TemporaryDirectory, or mkstemp with restrictive permissions and reliable cleanup.

Partial control mappings: EXEC-04. A mapping does not satisfy the full control.

Primary control source: MCP-ROOTS / JOINT-DEPLOY

Rule technical reference: TECH-PYTHON-SECURITY

Explore: invscan --explain-scan AI016

Run: invscan TARGET --scans AI016

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI017

JWT verification explicitly bypassed

AI017 / HIGH / authentication

WHAT THE DETERMINISTIC RULE DETECTS

Recognized Python JWT decode verification bypasses and JavaScript algorithms containing none.

ALGORITHM + SUPPORTED PATHS

JWT decode and literal algorithm/verification checks. Profiles: python_ast, javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

MCP or agent gateways may use token claims to select tools, users and tenants. An unverified token can let caller-controlled claims impersonate authority across those boundaries.

CAN MISS OR MISCLASSIFY

Does not establish a complete issuer/audience/scope policy or that unverified claims reach authorization.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Verify signature, algorithm allowlist, issuer, audience, expiry, and scopes before using claims for authorization.

Partial control mappings: AUTH-03. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH-SEC

Thematic alignment: CIS-MCP-2026 / OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY

Explore: invscan --explain-scan AI017

Run: invscan TARGET --scans AI017

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI018

MCP package runner resolves an unpinned artifact

AI018 / MEDIUM / supply chain

WHAT THE DETERMINISTIC RULE DETECTS

MCP launch configurations using supported ephemeral package runners without an exact package version.

ALGORITHM + SUPPORTED PATHS

Structured MCP command/argument package-version check. Profiles: json_structured. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An MCP client often launches a package with its local user permissions and injected credentials. Unpinned npx/uvx resolution can replace the tool implementation between launches without a reviewed configuration change.

CAN MISS OR MISCLASSIFY

Does not resolve the registry, verify provenance or inspect transitive dependencies; non-JSON launch formats may not receive this predicate.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Pin an exact audited package version and lock or verify transitive artifacts. Prefer preinstalled verified tools in a controlled environment.

Partial control mappings: MCP-09, SUP-01. A mapping does not satisfy the full control.

Primary control source: MCP-SECURITY / OWASP-MCP-CS / NIST-SSDF

Thematic alignment: OWASP-AISVS-C10 / NIST-SSDF-AI / AIUC1-Q3-2026

Rule technical reference: MCP-SECURITY / CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI018

Run: invscan TARGET --scans AI018

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI019

Remote download executed directly by a shell

AI019 / HIGH / supply chain

WHAT THE DETERMINISTIC RULE DETECTS

curl/wget output directly piped to a shell or used as shell -c command text in supported shell/configuration contexts.

ALGORITHM + SUPPORTED PATHS

Bounded shell token/command-substitution and pipeline patterns. Profiles: configuration_lexical, generic_text. Image evidence contexts: final_filesystem, build_history.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A tool bootstrap or agent image build that pipes downloaded text into a shell or uses it directly as shell command text lets the remote response choose commands under the build/runtime identity, potentially reaching deployment credentials.

CAN MISS OR MISCLASSIFY

Not a complete shell interpreter; complex expansion, wrappers, runtime fetch destinations and remote artifact contents are not resolved.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Fetch a versioned artifact, verify an independently trusted digest or signature, review it, and execute it under least privilege.

Partial control mappings: SUP-03. A mapping does not satisfy the full control.

Primary control source: NSA-DATA / NIST-SSDF

Thematic alignment: CSA-AICM / CSA-MAESTRO / NIST-SSDF-AI / OWASP-AISVS-C10 / SLSA-12 / OPENSSEF-MODEL-SIGNING

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI019

Run: invscan TARGET --scans AI019

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI020

GitHub Action reference is not pinned to a commit

AI020 / MEDIUM / supply chain

WHAT THE DETERMINISTIC RULE DETECTS

Remote action uses references lacking a reviewed full commit SHA in recognized workflow files.

ALGORITHM + SUPPORTED PATHS

GitHub workflow uses-reference pattern and full-SHA check. Profiles: configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent/MCP build and release workflows may possess signing, registry or deployment credentials. A mutable Action reference allows workflow code to change without a repository workflow edit.

CAN MISS OR MISCLASSIFY

Pinning alone does not establish the action is safe, signed, reviewed or minimally privileged.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Pin remote actions to a reviewed full commit SHA and use dependency automation to propose updates.

Partial control mappings: SUP-04. A mapping does not satisfy the full control.

Primary control source: NIST-SSDF

Thematic alignment: NIST-SSDF-AI / CIS-MCP-2026 / OPENSSF-BASELINE-202608 / SLSA-12

Rule technical reference: TECH-GITHUB-ACTIONS

Explore: invscan --explain-scan AI020

Run: invscan TARGET --scans AI020

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI021

Container explicitly runs as root

AI021 / MEDIUM / sandboxing

WHAT THE DETERMINISTIC RULE DETECTS

Explicit/inherited literal root USER in the default final Dockerfile stage; built-image configured user/default UID 0 is separately checked.

ALGORITHM + SUPPORTED PATHS

Dockerfile final-stage user tracking plus built-image user inspection. Profiles: configuration_lexical. Image evidence contexts: final_filesystem, runtime_configuration.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An injected or overbroad agent/MCP tool action inherits the container process identity. Root can increase access to mounted data and the impact of an isolation failure, although container root is not automatically host root.

CAN MISS OR MISCLASSIFY

External Dockerfile base defaults, variables and selected build targets may be unresolved; deployed user overrides require runtime configuration.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use a dedicated nonroot runtime user and minimal capabilities; verify the final build stage and deployment security context.

Partial control mappings: SUP-05. A mapping does not satisfy the full control.

Primary control source: JOINT-DEPLOY

Thematic alignment: CSA-CCM / CSA-MAESTRO

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI021

Run: invscan TARGET --scans AI021

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI022

Privileged container or privilege escalation enabled

AI022 / HIGH / sandboxing

WHAT THE DETERMINISTIC RULE DETECTS

Explicit privileged or allowPrivilegeEscalation settings in supported configuration.

ALGORITHM + SUPPORTED PATHS

Exact privilege-setting predicates. Profiles: json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent-selected tool actions in a privileged container can reach host-facing capabilities beyond the intended task. This can turn a tool compromise into a wider infrastructure incident.

CAN MISS OR MISCLASSIFY

Not a complete container capability or admission-policy audit; missing settings are not treated as proof of isolation.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Disable privileged mode and privilege escalation, drop capabilities, and expose only narrowly required devices or operations.

Partial control mappings: SUP-05. A mapping does not satisfy the full control.

Primary control source: JOINT-DEPLOY

Thematic alignment: CSA-CCM / CSA-MAESTRO

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI022

Run: invscan TARGET --scans AI022

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI023

Container runtime socket exposed

AI023 / HIGH / sandboxing

WHAT THE DETERMINISTIC RULE DETECTS

Recognized Docker/containerd runtime socket paths in supported configuration.

ALGORITHM + SUPPORTED PATHS

Known runtime-socket path matching. Profiles: json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A model-influenced tool with Docker/containerd API access can control workloads and potentially host-mounted files. A socket mount bypasses many restrictions otherwise applied to the agent container.

CAN MISS OR MISCLASSIFY

Presence does not prove the mount is active or writable; read-only filesystem mounts do not necessarily restrict socket API operations.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Remove runtime socket access or use a tightly scoped broker outside the agent sandbox. Read-only mounts do not necessarily make socket APIs read-only.

Partial control mappings: SUP-05. A mapping does not satisfy the full control.

Primary control source: JOINT-DEPLOY

Thematic alignment: CSA-CCM / CSA-MAESTRO

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI023

Run: invscan TARGET --scans AI023

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI024

Container image is not digest pinned

AI024 / LOW / supply chain

WHAT THE DETERMINISTIC RULE DETECTS

Mutable Dockerfile FROM or supported deployment image references rather than content digests.

ALGORITHM + SUPPORTED PATHS

Container reference parsing and digest-presence check. Profiles: json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent/MCP deployments may execute a different tool stack when a mutable image tag moves. That changes the code holding credentials and enforcing tool policy without a matching application source change.

CAN MISS OR MISCLASSIFY

No registry lookup, signature verification, CVE database or guarantee of safe pinned content.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Pin approved image digests, track provenance and SBOMs, and update through reviewed vulnerability-remediation workflows.

Partial control mappings: SUP-01, SUP-03. A mapping does not satisfy the full control.

Primary control source: NIST-SSDF / NSA-DATA

Thematic alignment: NIST-SSDF-AI / AIUC1-Q3-2026 / CSA-AICM / CSA-MAESTRO / OWASP-AISVS-C10 / SLSA-12 / OPENSSEF-MODEL-SIGNING

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI024

Run: invscan TARGET --scans AI024

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI025

Direct dependency is not exactly pinned

AI025 / LOW / supply chain

WHAT THE DETERMINISTIC RULE DETECTS

Recognized package.json dependency declarations and requirements-style entries permitting version movement; a local '.' self-install is excluded.

ALGORITHM + SUPPORTED PATHS

Direct manifest dependency version-shape checks. Profiles: json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent SDKs, MCP libraries and tool dependencies execute inside the application trust boundary. Uncontrolled dependency changes can alter prompt handling, authentication or tool behavior with the same source revision.

CAN MISS OR MISCLASSIFY

Lockfile inventory does not resolve the effective build or transitive graph; this is not a CVE, dependency-confusion or integrity assessment.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use a committed verified lockfile or exact pins and hashes, scan dependencies for known vulnerabilities, and review updates.

Partial control mappings: SUP-01. A mapping does not satisfy the full control.

Primary control source: NIST-SSDF

Thematic alignment: NIST-SSDF-AI / AIUC1-Q3-2026

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI025

Run: invscan TARGET --scans AI025

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI026

Authentication explicitly disabled

AI026 / HIGH / authentication

WHAT THE DETERMINISTIC RULE DETECTS

Supported authentication fields with the actual false value in code/configuration, including bounded supported YAML aliases.

ALGORITHM + SUPPORTED PATHS

Explicit authentication-disable field matching. Profiles: `python_ast`, `javascript_lexical`, `json_structured`, `configuration_lexical`. Image evidence contexts: `final_filesystem`.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A reachable MCP or agent service with disabled authentication can expose its tool authority to callers outside the intended user session. Direct local stdio instead relies on a process/OS boundary that must be assessed separately.

CAN MISS OR MISCLASSIFY

Local stdio may rely on the OS boundary; absent authentication code or custom middleware is not automatically classified.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

For remote protected resources, require authenticated clients and scoped per-tool authorization. Document local-only trust boundaries and protect launch configuration.

Partial control mappings: AUTH-01. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH

Thematic alignment: CSA-AGENT-IAM / CIS-MCP-2026 / OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY

Explore: `invscan --explain-scan AI026`

Run: `invscan TARGET --scans AI026`

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI027

Wildcard tool or permission authorization

AI027 / HIGH / authorization

WHAT THE DETERMINISTIC RULE DETECTS

Supported tool, permission and resource allowlists granting '*', including bounded supported YAML aliases.

ALGORITHM + SUPPORTED PATHS

Literal tool/resource wildcard matching. Profiles: python_ast, javascript_lexical, json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent with all-tools or wildcard permissions can turn a misinterpreted task or injected instruction into unrelated actions. Model-side tool selection does not limit a direct client calling the same server.

CAN MISS OR MISCLASSIFY

Does not enumerate effective IAM permissions or validate argument/resource policy and approval at runtime.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Replace wildcard grants with named tools and constrained resources; enforce authorization server-side for each call and require approval for sensitive effects.

Partial control mappings: AUTH-02, AGT-01. A mapping does not satisfy the full control.

Primary control source: [JOINT-AGENTIC](#) / [MCP-TOOLS](#) / [ASD-HARNESS](#)

Thematic alignment: [CSA-AICM](#) / [CSA-AGENT-IAM](#) / [CSA-SCOPING](#) / [CIS-MCP-2026](#) / [OWASP-AISVS-C9](#) / [OWASP-AISVS-C10](#) / [AIUC1-Q3-2026](#)

Rule technical reference: [MCP-SECURITY](#) / [OWASP-LLM2026](#)

Explore: `invscan --explain-scan AI027`

Run: `invscan TARGET --scans AI027`

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI028

MCP token passthrough enabled

AI028 / HIGH / authentication

WHAT THE DETERMINISTIC RULE DETECTS

Supported token-passthrough/forwarding settings explicitly enabled.

ALGORITHM + SUPPORTED PATHS

Explicit token-forwarding field matching. Profiles: `python_ast`, `javascript_lexical`, `json_structured`, `configuration_lexical`. Image evidence contexts: `final_filesystem`.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A server that forwards a caller token to another service can act as a confused intermediary, bypassing audience, consent and scope boundaries between the MCP resource and downstream tools.

CAN MISS OR MISCLASSIFY

Does not trace actual downstream token audiences, exchange flows or proxy behavior.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Validate tokens for this MCP resource and obtain separate audience-bound downstream credentials through an appropriate authorization flow.

Partial control mappings: AUTH-04. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH / MCP-SECURITY

Thematic alignment: CIS-MCP-2026 / OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY

Explore: `invscan --explain-scan AI028`

Run: `invscan TARGET --scans AI028`

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI029

Remote MCP URL uses plaintext HTTP

AI029 / HIGH / transport security

WHAT THE DETERMINISTIC RULE DETECTS

Nonloopback plaintext HTTP endpoints in recognized MCP configuration.

ALGORITHM + SUPPORTED PATHS

Structured MCP URL scheme and loopback classification. Profiles: json_structured. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Remote MCP requests can contain access tokens, task context and tool output. Plain HTTP exposes those bytes and allows modification by parties able to observe or influence the network path.

CAN MISS OR MISCLASSIFY

Transport encryption outside the application, DNS trust and deployment-specific tunnels are unverified; other endpoint formats may be missed.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use HTTPS with certificate verification for remote MCP endpoints. Reserve plaintext HTTP for explicitly controlled local development or a documented protected transport.

Partial control mappings: MCP-01. A mapping does not satisfy the full control.

Primary control source: MCP-HTTP

Thematic alignment: OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY

Explore: invscan --explain-scan AI029

Run: invscan TARGET --scans AI029

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI030

Browser-side API credential access explicitly enabled

AI030 / MEDIUM / secrets

WHAT THE DETERMINISTIC RULE DETECTS

dangerouslyAllowBrowser enabled in supported JavaScript/JSON fields.

ALGORITHM + SUPPORTED PATHS

Explicit browser-SDK opt-in field matching. Profiles: javascript_lexical, json_structured. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A browser-facing agent may expose model or tool credentials through a bundled SDK configuration. Any page user or compromised script could then call the API outside the intended agent task and server authorization.

CAN MISS OR MISCLASSIFY

Does not establish that a privileged long-lived credential actually reaches a browser bundle.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Keep durable API credentials on a trusted server, or use narrowly scoped short-lived client tokens designed for the browser.

Partial control mappings: AUTH-05, DATA-01. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH-SEC / JOINT-DEPLOY

Thematic alignment: CSA-AGENT-IAM / CIS-AGENTS-2026 / CIS-MCP-2026 / OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI030

Run: invscan TARGET --scans AI030

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI031

Agent approval or sandbox safeguard explicitly bypassed

AI031 / HIGH / agent permissions

WHAT THE DETERMINISTIC RULE DETECTS

Supported unrestricted agent approval/sandbox configuration and recognized dangerous CLI flags in applicable contexts.

ALGORITHM + SUPPORTED PATHS

Known bypass flags and typed unrestricted-approval settings. Profiles: python_ast, javascript_lexical, json_structured, configuration_lexical, generic_text. Image evidence contexts: final_filesystem, runtime_configuration, build_history.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Disabling agent approval or sandbox safeguards lets model-driven actions inherit the full process authority. Prompt injection or a mistaken plan can then modify files, send data or invoke credentials without the intended review boundary.

CAN MISS OR MISCLASSIFY

Cannot determine effective process permissions, external approval enforcement or whether the command actually runs.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use bounded tool permissions and an isolated runtime; require deliberate approval for destructive, external, financial, or credential-bearing actions.

Partial control mappings: MCP-09, AGT-01, AGT-02, SUP-05. A mapping does not satisfy the full control.

Primary control source: MCP-SECURITY / OWASP-MCP-CS / ASD-HARNESS / OWASP-AGENT-CS / JOINT-DEPLOY

Thematic alignment: OWASP-AISVS-C10 / CIS-MCP-2026 / OWASP-AISVS-C9 / AIUC1-Q3-2026 / CSA-SCOPING / CSA-CCM / CSA-MAESTRO

Rule technical reference: OWASP-LLM2026 / MCP-SECURITY

Explore: invscan --explain-scan AI031

Run: invscan TARGET --scans AI031

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI032

External data inserted into privileged LLM instructions

AI032 / MEDIUM / prompt injection

WHAT THE DETERMINISTIC RULE DETECTS

Supported system/developer message objects with obviously external content.

ALGORITHM + SUPPORTED PATHS

Privileged message-role plus local external-input matching. Profiles: python_ast, javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

External text inserted into a system/developer message is promoted into the same channel as trusted agent policy. A retrieved document or tool response may then steer planning and sensitive tool arguments.

CAN MISS OR MISCLASSIFY

Does not detect every prompt injection, prove a model follows it or trace a prompt assembled across functions/services.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Keep trusted instructions separate from untrusted content, label provenance, enforce tool policy outside the model, and test indirect prompt injection and sensitive-action approval.

Partial control mappings: AGT-03. A mapping does not satisfy the full control.

Primary control source: OWASP-LLM2026

Rule technical reference: OWASP-LLM2026

Explore: invscan --explain-scan AI032

Run: invscan TARGET --scans AI032

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI033

Potential secret included in logging call

AI033 / MEDIUM / data protection

WHAT THE DETERMINISTIC RULE DETECTS

Supported logging calls directly referencing credential-named variables, attributes or fields.

ALGORITHM + SUPPORTED PATHS

Logging sink and credential-name matching. Profiles: python_ast, javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent traces and MCP logs commonly cross observability vendors, support systems and retention tiers. Logging a credential widens who can use the underlying model or tool authority beyond its intended worker.

CAN MISS OR MISCLASSIFY

Variable names can be misleading; sanitizers, custom loggers, downstream formatters and unusual secret names may be missed.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Log nonsecret identifiers and outcomes; redact credentials before formatting and restrict log access and retention.

Partial control mappings: DATA-03. A mapping does not satisfy the full control.

Primary control source: OWASP-MCP10

Thematic alignment: CIS-AGENTS-2026 / CIS-MCP-2026 / AIUC1-Q3-2026

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI033

Run: invscan TARGET --scans AI033

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI034

Credential-like value embedded in URL query

AI034 / HIGH / data protection

WHAT THE DETERMINISTIC RULE DETECTS

URLs embedding a nonplaceholder credential-like query value.

ALGORITHM + SUPPORTED PATHS

URL query credential-name/value patterns. Profiles: generic_text. Image evidence contexts: final_filesystem, runtime_configuration, retained_layer, build_history.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent and MCP connectors often record request URLs in traces, tool results or browser history. A credential in the query string can be copied to systems that were never authorized to use the downstream service.

CAN MISS OR MISCLASSIFY

Cannot establish live credential validity, deployed URL use or all custom query parameter names.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use an appropriate authorization header or short-lived scoped credential; redact existing logs and rotate real exposed credentials.

Partial control mappings: AUTH-05, DATA-01. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH-SEC / JOINT-DEPLOY

Thematic alignment: CSA-AGENT-IAM / CIS-AGENTS-2026 / CIS-MCP-2026 / OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: MCP-SECURITY

Explore: invscan --explain-scan AI034

Run: invscan TARGET --scans AI034

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI035

Unsafe model checkpoint deserialization

AI035 / HIGH / model supply chain

WHAT THE DETERMINISTIC RULE DETECTS

`torch.load(weights_only=False)` and `recognized load_model(..., safe_mode=False)`.

ALGORITHM + SUPPORTED PATHS

Explicit model-loader safety opt-out checks. Profiles: `python_ast`. Image evidence contexts: `final_filesystem`.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Model loaders may run in agent infrastructure with access to prompts, datasets and deployment credentials. A malicious executable checkpoint can run producer-controlled code when loaded, before inference or tool policy applies.

CAN MISS OR MISCLASSIFY

Does not inspect model bytes, evaluate architecture behavior or infer unsafe defaults across every framework/version.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use safe tensor formats or restricted weights-only loading, verify model provenance and integrity, and isolate model conversion workflows.

Partial control mappings: EXEC-06, SUP-03. A mapping does not satisfy the full control.

Primary control source: [NIST-SSDF](#) / [NSA-DATA](#)

Thematic alignment: [CSA-AICM](#) / [CSA-MAESTRO](#) / [NIST-SSDF-AI](#) / [OWASP-AISVS-C10](#) / [SLSA-12](#) / [OPENSSF-MODEL-SIGNING](#)

Rule technical reference: [TECH-PYTORCH-SERIAL](#) / [CISA-SECURE-BY-DESIGN](#)

Explore: `invscan --explain-scan AI035`

Run: `invscan TARGET --scans AI035`

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI036

SQL query constructed with interpolation

AI036 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Supported database execution calls with interpolated or concatenated query text.

ALGORITHM + SUPPORTED PATHS

SQL execution sink plus interpolation/concatenation tracking. Profiles: python_ast, javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A database MCP tool may interpolate model-selected filters or generated SQL. If external text changes SQL structure, the tool can access or modify data beyond the intended user request and tenant.

CAN MISS OR MISCLASSIFY

Dynamic SQL identifiers may be intentionally allowlisted; cross-function parameterization and database-specific wrappers need review.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use bound query parameters for values and an explicit allowlist for dynamic SQL identifiers or operations.

Partial control mappings: EXEC-03. A mapping does not satisfy the full control.

Primary control source: OWASP-OUTPUT2025

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI036

Run: invscan TARGET --scans AI036

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI037

Archive extraction without an explicit safe filter

AI037 / MEDIUM / filesystem

WHAT THE DETERMINISTIC RULE DETECTS

Recognized tar extract/extractall calls without filter='data' or selecting a fully trusted filter.

ALGORITHM + SUPPORTED PATHS

Tar object/API tracking and explicit extraction-filter check. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agents may unpack repository snapshots, uploaded datasets or MCP tool artifacts. Archive member paths and links can otherwise overwrite files used by the agent or expose material outside its workspace.

CAN MISS OR MISCLASSIFY

Runtime-version defaults, custom safe filters and member-validation wrappers may need review; this does not audit every archive format.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use filter='data' on supported runtimes, validate members and resource limits, and extract into an isolated directory with no sensitive files.

Partial control mappings: EXEC-04. A mapping does not satisfy the full control.

Primary control source: MCP-ROOTS / JOINT-DEPLOY

Rule technical reference: TECH-PYTHON-TAR

Explore: invscan --explain-scan AI037

Run: invscan TARGET --scans AI037

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI038

Cryptographically weak randomness used for a security value

AI038 / MEDIUM / authentication

WHAT THE DETERMINISTIC RULE DETECTS

Credential-, nonce-, token- or secret-named assignments using the general-purpose random module.

ALGORITHM + SUPPORTED PATHS

Security-named assignment plus random API matching. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Predictable agent session IDs, MCP access tokens or approval nonces may let another caller guess authority or replay an action. A general-purpose random generator does not provide that security guarantee.

CAN MISS OR MISCLASSIFY

Name heuristics do not prove security use or entropy requirements; custom weak generators and JS randomness are not comprehensively analyzed.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use the secrets module or a cryptographically secure runtime API with sufficient entropy for security values.

Partial control mappings: AUTH-06, MCP-05. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH-SEC / MCP-AUTH / MCP-TOOLS / MCP-SECURITY

Rule technical reference: TECH-PYTHON-SECURITY

Explore: invscan --explain-scan AI038

Run: invscan TARGET --scans AI038

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI039

Server-Side Template rendered from a dynamic template

AI039 / HIGH / tool execution

WHAT THE DETERMINISTIC RULE DETECTS

Dynamic Flask render_template_string or Jinja2 Template source.

ALGORITHM + SUPPORTED PATHS

Recognized template-source sink and constant exclusion. Profiles: python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An agent or MCP tool may render generated reports, emails or pages. If model/user content becomes server-side template source, template expressions can access server objects or operations under the tool identity.

CAN MISS OR MISCLASSIFY

Template contents, sandbox configuration, wrapper calls and actual attacker control require context.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use a fixed trusted template and pass untrusted content only as escaped data; sandbox and limit any intentional template authoring.

Partial control mappings: EXEC-07. A mapping does not satisfy the full control.

Primary control source: OWASP-OUTPUT2025

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI039

Run: invscan TARGET --scans AI039

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI040

Unsafe HTML rendering boundary

AI040 / MEDIUM / output handling

WHAT THE DETERMINISTIC RULE DETECTS

Supported innerHTML, insertAdjacentHTML and dangerouslySetInnerHTML dynamic assignments.

ALGORITHM + SUPPORTED PATHS

Raw-HTML sink and dynamic-value matching. Profiles: javascript_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent messages and MCP results may contain attacker-origin content. Passing that output to innerHTML, dangerouslySetInnerHTML or a similar raw sink can execute browser markup with access to the application session.

CAN MISS OR MISCLASSIFY

Does not validate a sanitizer, framework escaping, browser CSP or DOM behavior at runtime.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Render untrusted output as text or sanitize with a maintained HTML policy before using a raw-HTML sink; apply a restrictive content security policy.

Partial control mappings: EXEC-07. A mapping does not satisfy the full control.

Primary control source: OWASP-OUTPUT2025

Rule technical reference: OWASP-LLM2026

Explore: invscan --explain-scan AI040

Run: invscan TARGET --scans AI040

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI041

MCP inspector authorization explicitly disabled

AI041 / HIGH / authentication

WHAT THE DETERMINISTIC RULE DETECTS

Recognized DANGEROUSLY_OMIT_AUTH assignments with a nonempty value; strings 'false' and '0' still disable the safeguard.

ALGORITHM + SUPPORTED PATHS

Inspector environment/configuration nonempty-value semantics. Profiles: python_ast, javascript_lexical, json_structured, configuration_lexical. Image evidence contexts: final_filesystem, runtime_configuration.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

MCP Inspector can reach connected servers and their tool permissions. A reachable unauthenticated inspector can expose that authority outside the intended developer session.

CAN MISS OR MISCLASSIFY

Does not determine Inspector reachability or runtime environment overrides; an empty string alone is not treated as enabled.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Retain inspector authentication and restrict its listeners and origins to a controlled development environment.

Partial control mappings: AUTH-01. A mapping does not satisfy the full control.

Primary control source: MCP-AUTH

Thematic alignment: CSA-AGENT-IAM / CIS-MCP-2026 / OWASP-AISVS-C10

Rule technical reference: MCP-SECURITY

Explore: invscan --explain-scan AI041

Run: invscan TARGET --scans AI041

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI042

Container uses host namespaces

AI042 / HIGH / sandboxing

WHAT THE DETERMINISTIC RULE DETECTS

hostPID/hostNetwork enabled or supported pid/network_mode values set to host.

ALGORITHM + SUPPORTED PATHS

Explicit host namespace setting checks. Profiles: json_structured, configuration_lexical. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Host PID or network namespace sharing lets an agent or MCP tool observe or interact with host-level processes or services beyond its isolated workload. Model-influenced actions may then reach assets outside the intended task sandbox.

CAN MISS OR MISCLASSIFY

Does not inventory all namespace, seccomp, capability or kernel-isolation behavior in a running deployment.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use isolated namespaces and explicit network rules; document and minimize any unavoidable host-level access.

Partial control mappings: SUP-05. A mapping does not satisfy the full control.

Primary control source: JOINT-DEPLOY

Thematic alignment: CSA-CCM / CSA-MAESTRO

Rule technical reference: CISA-SECURE-BY-DESIGN

Explore: invscan --explain-scan AI042

Run: invscan TARGET --scans AI042

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI043

Instruction-hierarchy override in agent-facing text

AI043 / HIGH / prompt injection

WHAT THE DETERMINISTIC RULE DETECTS

Recognized instructions and literal tool/schema descriptions asking to override trusted instructions; selected Spanish/French/German override forms; paired authority invalidation/replacement; and a supported conditional conflict resolved in favor of the current tool or skill.

ALGORITHM + SUPPORTED PATHS

Bounded directive predicates and conflicting authority claims. Profiles: instruction_text, tool_metadata. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

A tool description or skill instruction can be inserted into model context before invocation; an override directive attempts to cross the boundary from external data into agent authority.

CAN MISS OR MISCLASSIFY

Limited lexical language forms are not translation coverage. Other languages, paraphrases, reordered authority claims, unsupported encodings and dynamic descriptions can evade detection. Evidence does not prove intent or successful injection.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Remove the override directive, keep tool metadata scoped to its function, and enforce instruction/data separation and sensitive-action policy outside the model. Test the exact text against the deployed agent with harmless canary actions.

Partial control mappings: MCP-03, AGT-03, AGT-06. A mapping does not satisfy the full control.

Primary control source: MCP-TOOLS / OWASP-MCP-CS / OWASP-LLM2026 / MITRE-ATLAS / NCSC-SECURE-AI / AGENT-SKILLS-SPEC

Thematic alignment: OWASP-AISVS-C10 / AIUC1-Q3-2026

Rule technical reference: MCP-TOOLS / AGENT-SKILLS-SPEC / OWASP-LLM2026

Explore: invscan --explain-scan AI043

Run: invscan TARGET --scans AI043

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI044

Sensitive-data transfer instruction

AI044 / HIGH / data protection

WHAT THE DETERMINISTIC RULE DETECTS

Recognized English transfer instructions naming credentials, sensitive files, a credential store or login/password vault; nearest-operative-read to pronoun/file/attachment references; and include/append bundle forms with an explicit URL/email destination in the bounded segment.

ALGORITHM + SUPPORTED PATHS

Sensitive-object, transfer-action, local negation and explicit-destination predicates. Profiles: instruction_text, tool_metadata. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

An instruction that directs an agent to gather a credential or sensitive file and send it to a URL/email can turn an otherwise legitimate tool into a disclosure path.

CAN MISS OR MISCLASSIFY

Local explicit transfer negation is respected where recognized. A later public-object retrieval does not inherit an earlier sensitive referent. Implicit destinations, multi-file staging and natural-language ambiguity remain unresolved; actual access, authorization, transfer and destination ownership are unverified.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Remove credential collection from tool descriptions and skill instructions. Use a scoped credential broker, allowlist outbound destinations, redact tool inputs and require approval displaying the actual recipient and data before any sensitive disclosure.

Partial control mappings: MCP-03, AGT-06, AGT-07. A mapping does not satisfy the full control.

Primary control source: MCP-TOOLS / OWASP-MCP-CS / MITRE-ATLAS / NCSC-SECURE-AI / AGENT-SKILLS-SPEC

Thematic alignment: OWASP-AISVS-C10 / AIUC1-Q3-2026

Rule technical reference: MCP-TOOLS / AGENT-SKILLS-SPEC / MCP-SECURITY

Explore: invscan --explain-scan AI044

Run: invscan TARGET --scans AI044

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI045

Covert action or approval-bypass instruction

AI045 / HIGH / agent permissions

WHAT THE DETERMINISTIC RULE DETECTS

Recognized instructions pairing an action with concealment from the user, explicitly erasing audit evidence to prevent user inspection, or directing approval/sandbox bypass.

ALGORITHM + SUPPORTED PATHS

Action/concealment, audit-erasure and approval-bypass predicates. Profiles: instruction_text, tool_metadata. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Concealment or approval-bypass text in skills/tool descriptions can induce the planner to exceed user-authorized actions under the agent or MCP process identity.

CAN MISS OR MISCLASSIFY

Cannot establish deceptive intent from all natural-language variations; legitimate quoted security examples are excluded only where the bounded parser recognizes them.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Delete the concealment or bypass directive. Bind approvals to the exact operation and arguments, enforce policy in the tool server, retain auditable execution records and isolate the process with minimum privileges.

Partial control mappings: AGT-02, AGT-06. A mapping does not satisfy the full control.

Primary control source: OWASP-AGENT-CS / MITRE-ATLAS / NCSC-SECURE-AI / AGENT-SKILLS-SPEC

Thematic alignment: CSA-SCOPING / OWASP-AISVS-C9 / AIUC1-Q3-2026

Rule technical reference: MCP-TOOLS / AGENT-SKILLS-SPEC / MCP-SECURITY

Explore: invscan --explain-scan AI045

Run: invscan TARGET --scans AI045

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI046

Read-only tool annotation conflicts with declared or observed writes

AI046 / MEDIUM / tool integrity

WHAT THE DETERMINISTIC RULE DETECTS

Literal readOnlyHint=true conflicting with a destructive instruction in the same description, or a top-level Python @tool handler containing an unambiguously resolved os/shutil write call or inline pathlib.Path write operation. Literal annotation dictionaries and ToolAnnotations calls are recognized.

ALGORITHM + SUPPORTED PATHS

Read-only annotation consistency with description or direct write witness. Profiles: tool_metadata, python_ast. Image evidence contexts: final_filesystem.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Agent and MCP clients may prioritize tools or reduce consent based on readOnlyHint. A destructive description or resolved direct write in a supported Python handler conflicts with that declared boundary; actual execution and authorization still require validation.

CAN MISS OR MISCLASSIFY

Python effects are bounded to 20000 inspected handler AST nodes; nested/class bindings leave gaps. Shadowed/conditional imports, uncalled nested bodies and recognized dead branches are excluded. Helper calls, variable-held Path objects, arbitrary writes, runtime registration, authorization and actual branch execution are not proven.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Inspect the implementation and correct its description and annotations together. Treat the tool as potentially mutating until verified, reapprove changed metadata and enforce write authorization independently of model decisions or annotation hints.

Partial control mappings: MCP-03, MCP-04. A mapping does not satisfy the full control.

Primary control source: MCP-TOOLS / OWASP-MCP-CS / OWASP-MCP10

Thematic alignment: OWASP-AISVS-C10

Rule technical reference: MCP-TOOLS / MCP-SECURITY

Explore: invscan --explain-scan AI046

Run: invscan TARGET --scans AI046

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

06 / EXACT SCAN CONTRACT / AI047

Filesystem permissions explicitly allow writes by everyone

AI047 / MEDIUM / filesystem

WHAT THE DETERMINISTIC RULE DETECTS

Recognized `os.chmod/fchmod/lchmod` and `pathlib.chmod/lchmod` calls whose literal, propagated integer, stat flag or bounded bitwise mode explicitly contains the other-write bit (0o002).

ALGORITHM + SUPPORTED PATHS

Resolved `chmod` API plus bounded integer/permission-bit evaluation. Profiles: `python_ast`. Image evidence contexts: `final_filesystem`.

WHY THIS MATTERS FOR AGENTS, MCP AND SKILLS

Another local identity may replace agent-generated code, tool artifacts, instructions or credentials when the shared filesystem permits writes by everyone. Broad permissions can defeat an otherwise correct tool authorization check.

CAN MISS OR MISCLASSIFY

Only explicit supported `chmod` calls are checked. Owner-only/group-only writes are not this rule. Platform behavior, ACLs, ownership, `umask`, reachability and live permissions remain unverified; a sticky bit does not erase the other-write signal.

OPTIONAL AI REVIEW

Can explain applicability and suggest follow-up evidence for detected patterns; cannot remove or downgrade a deterministic finding. Full mode reviews selected active controls even with zero rule matches. Optional bounded investigation can request additional ranges from the captured, hash-verified manifest snapshot and seek counterevidence before proposing a gap. It cannot execute target tools or validate runtime behavior.

HOW TO FIX OR VALIDATE

Use owner-only permissions (typically 0o600 for private files and 0o700 for private directories), or a narrowly controlled shared group where required. Verify access from the actual worker and an unrelated identity; avoid granting 0o777 merely to fix container UID mismatches.

Partial control mappings: EXEC-04. A mapping does not satisfy the full control.

Primary control source: MCP-ROOTS / JOINT-DEPLOY

Rule technical reference: TECH-PYTHON-CHMOD / CWE-732

Explore: `invscan --explain-scan AI047`

Run: `invscan TARGET --scans AI047`

Full per-check requirements and runtime/human evidence remain in the mapped control cards. Model interpretation is advisory; citations confirm quoted text, not correctness. Nothing on this page establishes malicious intent or exploitability.

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

JOINT-AGENTIC**Careful adoption of agentic AI services**

ASD ACSC, CISA, NSA, CCCS, NCSC-NZ, NCSC-UK / Government guidance / Joint guidance / 2026-05-01 / Accessed 2026-09-19

Agent privileges, identity, trust boundaries, autonomy, oversight, supply chain and accountable operation. **Limit:** ASD publication date; NSA announced it on April 30. Recommendations require system-specific implementation and evidence; this is not an executable benchmark.

www.cyber.gov.au/business-government/secure-design/artificial-intelligence/careful-adoption-of-a...

NSA-AGENTIC**NSA announcement: Careful Adoption of Agentic AI Services**

NSA / Government announcement / Joint guidance release / 2026-04-30 / Accessed 2026-09-19

Confirms NSA participation, publication timing and joint authorship of agentic AI guidance. **Limit:** Announcement supports provenance; use the complete joint guidance for substantive control interpretation.

www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4475134/nsa-joins-th...

CISA-AGENTIC**CISA and Partners Release Guidance on Adopting Agentic AI Services**

CISA / Government announcement / Official CISA bulletin / 2026-05-01 / Accessed 2026-09-19

Confirms CISA participation and recommendations to bound autonomy and strengthen identity and oversight. **Limit:** Official bulletin hosted by GovDelivery. This is a release notice, not a separately scored framework.

content.govdelivery.com/accounts/USDHSCISA/bulletins/41544ff

ASD-HARNESS**Agentic AI harnesses: The layer above the model**

ASD ACSC / Government guidance / Published guidance / 2026-09-11 / Accessed 2026-09-19

Security responsibilities of orchestration, context, tool execution, memory, permissions and operational controls. **Limit:** ASD publication; do not attribute joint authorship to NSA or CISA. A secure model alone does not establish harness security.

www.cyber.gov.au/business-government/secure-design/artificial-intelligence/agentic-ai-harnesses

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

JOINT-DEPLOY

Deploying AI Systems Securely: Best Practices for Deploying Secure and Resilient AI Systems

NSA, CISA and international partners; ASD host / Government guidance / 1.0 / 2024-04-16 / Accessed 2026-09-19

Infrastructure hardening, deployment controls, data protection, monitoring and secure AI operation. **Limit:** ASD page date. Focuses on operating externally developed AI, especially on-premises or private cloud; adapt infrastructure checks to hosted API usage.

www.cyber.gov.au/business-government/secure-design/artificial-intelligence/deploying-ai-systems-...

NSA-DATA

AI Data Security: Best Practices for Securing Data Used to Train & Operate AI Systems (NSA announcement)

NSA, CISA, FBI and international partners / Government guidance announcement / Joint Cybersecurity Information Sheet / 2025-05-22 / Accessed 2026-09-19

Data provenance, integrity, protection and lifecycle changes affecting training and operational data. **Limit:** The linked announcement points to the full information sheet. Local source review cannot establish provider practices or retrieval-corpus integrity.

www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4192332/nsas-aisc-re...

NCSC-SECURE-AI

Guidelines for secure AI system development

NCSC-UK, CISA and international partners / Government guidance / 1.0 / 2023-11-27 / Accessed 2026-09-19

Secure design, development, deployment, operation and maintenance across the AI lifecycle. **Limit:** Lifecycle guidance requires implementation evidence; it does not prescribe an agent/MCP scanner or certify code.

www.ncsc.gov.uk/collection/guidelines-secure-ai-system-development

CISA-JCDC

JCDC AI Cybersecurity Collaboration Playbook and Fact Sheet (release notice)

CISA / Government operational guidance / 2025 release / 2025-01-14 / Accessed 2026-09-19

Incident collaboration, voluntary sharing processes and coordinated AI cybersecurity response. **Limit:** Operational playbook, not a source-code scanning standard or a mandate to disclose all findings.

www.cisa.gov/news-events/alerts/2025/01/14/cisa-releases-jcdc-ai-cybersecurity-collaboration-pla...

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

NIST-RMF

Artificial Intelligence Risk Management Framework (AI RMF 1.0)

NIST / Risk-management framework / AI 100-1; 1.0 / 2023-01-26 / Accessed 2026-09-19

Risk ownership, system context, measurement and risk treatment. **Limit:** Voluntary framework, not a universal technical pass/fail checklist; project mappings select security-relevant practices.

www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10

NIST-RMF-PROGRAM

AI Risk Management Framework: program status

NIST / Program reference / Living page / Living page / Accessed 2026-09-19

Current program and revision information for AI RMF. **Limit:** A changing program page is not a released replacement for the cited AI RMF 1.0 publication.

www.nist.gov/itl/ai-risk-management-framework

NIST-GENAI

Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile

NIST / Risk-management profile / AI 600-1 / 2024-07 / Accessed 2026-09-19

GenAI information security, privacy, data, evaluation and human interaction risks. **Limit:** Broader than agent/MCP security; selected mappings do not cover the whole profile. NIST records public release on July 26, 2024.

nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

NIST-RMF-RESOURCES

AI Risk Management Framework resources

NIST / Program reference / Living page / Living page / Accessed 2026-09-19

Publication context and release-date provenance for AI RMF resources and the GenAI Profile. **Limit:** Reference index, not a separate control framework; link to each actual publication when interpreting its requirements.

www.nist.gov/itl/ai-risk-management-framework/ai-risk-management-framework-resources

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

NIST-AML

Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations

NIST / Threat taxonomy / AI 100-2e2025 / 2025-03-24 / Accessed 2026-09-19

Attacker capabilities, goals and attack classes including poisoning, privacy and prompt injection. **Limit:** Taxonomy is not conformance testing. The official record has a June 3, 2025 potential-updates note; check it before relying on precise identifiers.

csrc.nist.gov/pubs/ai/100/2/e2025/final

NIST-SSDF

Secure Software Development Framework (SSDF) Version 1.1

NIST / Secure-development framework / SP 800-218; 1.1 / 2022-02-03 / Accessed 2026-09-19

Secure engineering, software integrity, vulnerability response and release lifecycle practices. **Limit:** Referenced for engineering context; this catalog is not an exhaustive SSDF compliance crosswalk.

csrc.nist.gov/pubs/sp/800/218/final

NIST-HIJACK-EVAL

Technical Blog: Strengthening AI Agent Hijacking Evaluations

NIST / Evaluation research / Technical blog / 2025-01-17 / Accessed 2026-09-19

Adaptive attacks, repeated attempts and scenario-specific evaluation of agent hijacking. **Limit:** Research methodology does not establish production assurance or a certification; report the tested harness and attack budget.

www.nist.gov/news-events/news/2025/01/technical-blog-strengthening-ai-agent-hijacking-evaluations

MITRE-ATLAS

ATLAS tactics, techniques, mitigations and case-study data

MITRE / Threat knowledge base / Living official repository / Living repository / Accessed 2026-09-19

Threat modeling, adversarial AI attack paths and security-test scenario design. **Limit:** Pin a content release and commit for exact IDs. Content and schema versions differ; this scanner does not calculate an ATLAS security score.

github.com/mitre-atlas/atlas-data

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

OWASP-AGENTIC

OWASP Top 10 for Agentic Applications for 2026

OWASP GenAI Security Project / Threat prioritization / 2026 edition / 2025-12-09 / Accessed 2026-09-19

Agent goals, tool use, identity, memory, code execution, communication and cascading failures. **Limit:** A prioritized risk list, not an executable benchmark. Date follows the page display; machine metadata uses the following UTC day.

genai.owasp.org/resource/owasp-top-10-for-agentic-applications-for-2026/

OWASP-LLM2026

OWASP GenAI LLM Top 10 2026

OWASP GenAI Security Project / Threat prioritization / 2026 edition / 2026-08-03 / Accessed 2026-09-19

Security risks of LLM-powered applications and their connected data and execution paths. **Limit:** Do not silently transfer category numbering from the 2025 edition. Date follows the displayed page rather than next-day UTC metadata.

genai.owasp.org/resource/owasp-genai-llm-top-10-2026/

OWASP-MCP10

OWASP MCP Top 10

OWASP / Threat prioritization / Living page; MCP01:2025-MCP10:2025 labels / Living page / Accessed 2026-09-19

MCP credentials, privilege scope, tool poisoning, supply chain, injection, audit and unapproved servers. **Limit:** Roadmap still describes beta/pilot even though the page also labels the project production. Treat maturity as mixed; it is not protocol conformance or benchmark completion.

owasp.org/projects/mcp-top-10

OWASP-MCP-CS

MCP Security Cheat Sheet

OWASP Cheat Sheet Series / Implementation guidance / Living document / Living document / Accessed 2026-09-19

Practical defenses for MCP identity, tools, data boundaries, operations and exposure. **Limit:** Reconcile version-sensitive recommendations against the actual deployed MCP revision; it is secondary to normative protocol requirements.

cheatsheetseries.owasp.org/cheatsheets/MCP_Security_Cheat_Sheet.html

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

OWASP-AGENT-CS

AI Agent Security Cheat Sheet

OWASP Cheat Sheet Series / Implementation guidance / Living document / Living document / Accessed 2026-09-19

Agent policies, isolation, context, tools and operational security testing. **Limit:** General implementation guidance requires architecture-specific interpretation and runtime evidence.

cheatsheetseries.owasp.org/cheatsheets/AI_Agent_Security_Cheat_Sheet.html

OWASP-OUTPUT2025

LLM05:2025 Improper Output Handling

OWASP GenAI Security Project / Threat category / LLM05:2025 / 2025 edition / Accessed 2026-09-19

Validate and encode model output before it reaches interpreters, commands, queries or presentation sinks. **Limit:** Explicit historical category; its identifier must not be described as the 2026 numbering.

genai.owasp.org/llmrisk/llm052025-improper-output-handling/

OWASP-ACS

Agent Control Standard (ACS)

OWASP GenAI Security Project / Emerging control architecture / Resource publication / 2026-09-01 / Accessed 2026-09-19

Runtime policy hooks and enforcement architecture for agent systems. **Limit:** Included as emerging architecture context; this scanner does not implement or claim conformance to ACS.

genai.owasp.org/resource/agent-control-standard-acs/

OWASP-CROSSWALK

GenAI Security Industry Framework Crosswalk

OWASP GenAI Security Project / Framework crosswalk / Resource publication / 2026-09-01 / Accessed 2026-09-19

Comparison context across AI security guidance and frameworks. **Limit:** Project mappings are independently scoped. Do not infer exhaustive adoption of this third-party crosswalk.

genai.owasp.org/resource/genai-security-industry-framework-crosswalk/

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

MCP-SPEC

Model Context Protocol specification

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Reference revision for client/server protocol and conditional interoperability requirements. **Limit:** Record the deployed revision. Versioned web pages can receive edits; pin upstream source for formal audit evidence.

modelcontextprotocol.io/specification/2026-07-28

MCP-AUTH

MCP: Authorization

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

HTTP authorization, protected-resource discovery, audience restrictions and OAuth responsibilities. **Limit:** HTTP authorization is optional in the protocol; protected applications still need suitable controls. Stdio uses a different credential model.

modelcontextprotocol.io/specification/2026-07-28/basic/authorization

MCP-AUTH-SEC

MCP: Authorization Security Considerations

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Token validation, PKCE, issuer binding, redirects and authorization threat defenses. **Limit:** Requirements are conditional on roles and deployed protocol version; decoding a token alone does not validate it.

modelcontextprotocol.io/specification/2026-07-28/basic/authorization/security-considerations

MCP-TRANSPORTS

MCP: Transport overview

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Transport selection, request metadata and compatibility across protocol revisions. **Limit:** The 2026-07-28 core differs from older session and initialize-handshake designs; do not label older code a violation without checking its advertised version.

modelcontextprotocol.io/specification/2026-07-28/basic/transports

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

MCP-HTTP

MCP: Streamable HTTP transport

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Origin validation, HTTP request behavior, streaming and transport-specific controls. **Limit:** Transport compliance does not establish downstream object authorization or safe tool behavior.

modelcontextprotocol.io/specification/2026-07-28/basic/transports/streamable-http

MCP-TOOLS

MCP: Tools

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Tool input/output schemas, identity, annotations, access controls and implementation responsibilities. **Limit:** Descriptions and annotations are untrusted metadata; they do not enforce read-only behavior or least privilege.

modelcontextprotocol.io/specification/2026-07-28/server/tools

MCP-SAMPLING

MCP: Sampling

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Host control over model sampling requests, tool access and shared context. **Limit:** Sampling permission does not automatically authorize server access to user context or downstream actions.

modelcontextprotocol.io/specification/2026-07-28/client/sampling

MCP-ELICITATION

MCP: Elicitation

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

User information requests, interaction modes, sensitive data and correlated responses. **Limit:** Validate supported modes and version-specific requirements; interaction affordances alone do not prevent phishing.

modelcontextprotocol.io/specification/2026-07-28/client/elicitation

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

MCP-ROOTS**MCP: Roots**

Model Context Protocol maintainers / Protocol specification / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Client-advertised filesystem roots and scope information supplied to servers. **Limit:** Advertised roots are not an operating-system sandbox; enforce filesystem access independently.

modelcontextprotocol.io/specification/2026-07-28/client/roots

MCP-SECURITY**MCP Security Best Practices**

Model Context Protocol maintainers / Protocol security guidance / 2026-07-28 / 2026-07-28 / Accessed 2026-09-19

Confused deputy, token forwarding, discovery SSRF, local launch, URL handling and registration trust. **Limit:** Companion guidance is not a production assurance result. Use this moved documentation path rather than the older specification path.

modelcontextprotocol.io/docs/2026-07-28/tutorials/security/security_best_practices

BENCH-AGENTDOJO**AgentDojo**

ETH Zurich / Invariant Labs / executable research benchmark / NeurIPS 2024; living repository / 2024 / Accessed 2026-09-19

Indirect prompt injection and legitimate task utility in tool workflows. **Limit:** Simulated suites and evolving API; adapt to production tools and policies. Not executed here.

github.com/ethz-spylab/agentdojo

BENCH-INJECAGENT**InjecAgent**

UIUC Kang Lab / executable research benchmark / 2024 research; living repository / 2024 / Accessed 2026-09-19

Malicious tool-result scenarios for tool-integrated agents. **Limit:** Fixed cases are not an adaptive attacker or production permission test. Not executed here.

github.com/uiuc-kang-lab/InjecAgent

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

BENCH-ASB

Agent Security Bench (ASB)

Agent Security Bench authors / agiresearch / executable research benchmark / ICLR 2025; living repository / 2025 / Accessed 2026-09-19

Multiple agent scenarios and attack/defense evaluation methods. **Limit:** Match harness, agent, and tools; published scores do not transfer automatically. Not executed here.

github.com/agiresearch/ASB

BENCH-MCPSECBENCH

MCPSecBench

AIS2Lab / executable research benchmark / 2025 research; living repository / 2025 / Accessed 2026-09-19

MCP security playground and attack scenarios across protocol surfaces. **Limit:** Environment/client-specific research workflows; not arbitrary-repository auditing or current-spec conformance. Not executed here.

github.com/AIS2Lab/MCPSecBench

CSA-AICM

AI Controls Matrix v1.1

Cloud Security Alliance / control framework and assessment questionnaire / 1.1 / 2026-06-22 / Accessed 2026-09-19

247 control objectives across 18 domains; AI service roles, lifecycle safeguards, and AI-CAIQ supplier assessment. **Limit:** Release page verified; complete workbook not reviewed or reproduced. Engineering alignment only; no exact AICM control mapping or certification claim.

cloudsecurityalliance.org/artifacts/ai-controls-matrix-v1-1

CSA-AICM-MACHINE

AICM Machine Readable Bundle (JSON/YAML/OSCAL)

Cloud Security Alliance / machine-readable framework distribution / Release page dated 2026-08-04; bundle not imported / 2026-08-04 / Accessed 2026-09-19

Publisher-provided machine-readable AICM controls, AI-CAIQ, implementation guidance, and mappings. **Limit:** Landing page verified only. Confirm exact downloadable version and licensing before importing into the scanner.

cloudsecurityalliance.org/artifacts/aicm-machine-readable-bundle-json-yaml-oscal

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

CSA-CCM

Cloud Controls Matrix and CAIQ v4.1

Cloud Security Alliance / cloud control framework and assessment questionnaire / 4.1 / 2026-01-27 / Accessed 2026-09-19

207 controls across 17 cloud security domains; infrastructure assurance and provider/customer responsibilities. **Limit:** Release page verified; no exhaustive control-ID mapping. Cloud assessment does not establish security of configured agent behavior.

cloudsecurityalliance.org/artifacts/cloud-controls-matrix-v4-1

CSA-MAESTRO

Agentic AI Threat Modeling Framework: MAESTRO

Cloud Security Alliance / threat-modeling framework / Publication dated 2025-02-06 / 2025-02-06 / Accessed 2026-09-19

Seven architectural layers and cross-layer attack paths for agentic systems. **Limit:** Analytical framework, not an executable benchmark or normative control catalog.

cloudsecurityalliance.org/blog/2025/02/06/agentic-ai-threat-modeling-framework-maestro

CSA-AGENT-IAM

Agentic AI Identity and Access Management: A New Approach

Cloud Security Alliance / architecture guidance / Publication dated 2025-08-18 / 2025-08-18 / Accessed 2026-09-19

Agent identity profiles, lifecycle governance, delegation, policy enforcement, and revocation. **Limit:** Proposed architecture; technology choices such as decentralized identifiers are not universal deployment requirements.

cloudsecurityalliance.org/artifacts/agentic-ai-identity-and-access-management-a-new-approach

CSA-SCOPING

Enhancing the Agentic AI Security Scoping Matrix: A Multi-Dimensional Approach

Cloud Security Alliance / risk-scoping guidance / Publication dated 2025-12-16 / 2025-12-16 / Accessed 2026-09-19

Access capability and risk exposure across identity, data, logging, agent controls, boundaries, and orchestration. **Limit:** CSA commentary extends an AWS framework. Scoping does not provide measured attack resistance.

cloudsecurityalliance.org/blog/2025/12/16/enhancing-the-agentic-ai-security-scoping-matrix-a-mul...

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

CSA-AISMM

AI Security Maturity Model (AISMM)

Cloud Security Alliance / security program maturity model / Publication dated 2026-05-19 / 2026-05-19 / Accessed 2026-09-19

Enterprise AI security program capabilities across self-hosted, PaaS, and API/SaaS deployments. **Limit:** Representative maturity indicators; not a complete checklist or a pass/fail assessment of one AI application.

cloudsecurityalliance.org/artifacts/ai-security-maturity-model

NIST-SSDF-AI

SP 800-218A: Secure Software Development Practices for Generative AI and Dual-Use Foundation Models

NIST / secure-development community profile / SP 800-218A, final / 2024-07-26 / Accessed 2026-09-19

AI-specific additions to SSDF 1.1 for model producers, system producers, and acquirers. **Limit:** Use with SP 800-218. Training and model-development controls require applicability decisions for API-only consumers.

csrc.nist.gov/pubs/sp/800/218/a/final

NIST-AGENT-IDENTITY-DRAFT

Accelerating the Adoption of Software and AI Agent Identity and Authorization Concept Paper

NIST National Cybersecurity Center of Excellence / draft concept paper / February 2026 draft / 2026-02-05 / Accessed 2026-09-19

Exploration of agent identification, authorization, and accountable access delegation using existing standards. **Limit:** Official page labels it Draft as accessed; not a completed practice guide, mandated standard, or executable benchmark.

www.nccoe.nist.gov/publications/other/accelerating-adoption-software-and-ai-agent-identity-and-a...

MITRE-ATLAS-202609

MITRE ATLAS content release v2026.09

MITRE / threat taxonomy / v2026.09 / 2026-09-15 / Accessed 2026-09-19

Versioned AI adversary techniques, mitigations, and case studies for threat modeling and test selection. **Limit:** A threat knowledge base, not a security score or evidence that mapped controls are effective.

github.com/mitre-atlas/atlas-data/releases/tag/v2026.09

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

CIS-CONTROLS-81

CIS Critical Security Controls v8.1

Center for Internet Security / control framework / v8.1 / 2024-06-24 / Accessed 2026-09-19

Enterprise security hygiene and implementation-group context for the deployed agent ecosystem. **Limit:** Organizational and deployment evidence required; no full CIS assessment performed.

www.cisecurity.org/controls/v8-1

CIS-AGENTS-2026

Artificial Intelligence (AI) Agents Companion Guide

Center for Internet Security / implementation guidance / v1.0 / CIS Controls v8.1 / 2026-04-20 / Accessed 2026-09-19

Agent runtimes, orchestration, tool use, memory, and operational ownership. **Limit:** Official 79-page PDF reviewed; independent checks and partial pointers only. CIS CC BY-NC-ND terms apply to original guide.

www.cisecurity.org/insights/white-papers/controls-v8-1-ai-agents-companion-guide

CIS-MCP-2026

Model Context Protocol (MCP) Companion Guide

Center for Internet Security / implementation guidance / v1.0 / CIS Controls v8.1 / 2026-04-20 / Accessed 2026-09-19

Security across MCP hosts, clients, servers, gateways, and connected systems. **Limit:** Official 82-page PDF reviewed; reconcile protocol-era details with deployed MCP. CIS CC BY-NC-ND terms apply to original guide.

www.cisecurity.org/insights/white-papers/controls-v8-1-model-context-protocol-companion-guide

CIS-AI-WORKBOOK

CIS Controls AI Security Guidance Workbook

Center for Internet Security / implementation workbook / Landing page describes workbook v8.1.2 / 2026-07-27 / Accessed 2026-09-19

Implementation tracking for LLM, agent, and MCP assets and owners. **Limit:** Public landing page reviewed; workbook cells not audited or imported. Workbook version is not a new overall CIS Controls release.

www.cisecurity.org/insights/white-papers/controls-v8-1-2-ai-security-guidance-workbook

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

CIS-MCP-BENCH-1

CIS MCP Server Benchmark 1.0.0 - official catalog listing

Center for Internet Security / configuration benchmark / MCP Server 1.0.0 / Publication date not stated in reviewed listing / Accessed 2026-09-19

Catalog identifies an MCP server configuration benchmark; select related platform benchmarks for actual deployment. **Limit:** Version/catalog verified only. Official full-benchmark download presents a registration form; full controls not reviewed, reproduced, or implemented.

www.cisecurity.org/cis-benchmarks

OWASP-AISVS-1

Artificial Intelligence Security Verification Standard (AISVS)

OWASP / verification standard / 1.0 / 2026-06 / Accessed 2026-09-19

Testable AI security requirements, including agent orchestration and MCP chapters. **Limit:** Selected chapters reviewed; not an exhaustive assessment. General application and infrastructure security must also be verified.

owasp.github.io/www-project-artificial-intelligence-security-verification-standard-aisvs-docs/

OWASP-AISVS-C9

AISVS 1.0 C9 - Orchestration and Agentic Security

OWASP / verification requirements / 1.0, C9 / 2026-06 / Accessed 2026-09-19

Agent execution budgets, approval, isolation, identity, delegation, and shutdown. **Limit:** Selected context mappings, not full AISVS conformance. Apply verification levels and preserve version-qualified identifiers.

github.com/OWASP/AISVS/blob/main/1.0/en/0x10-C09-Orchestration-and-Agentic-Action.md

OWASP-AISVS-C10

AISVS 1.0 C10 - Model Context Protocol Security

OWASP / verification requirements / 1.0, C10 / 2026-06 / Accessed 2026-09-19

MCP integrity, authorization, transport, schemas, and tool-change review. **Limit:** Application policy may exceed protocol requirements. Legacy initialize/session references require deployed-version interpretation; no full conformity claimed.

github.com/OWASP/AISVS/blob/main/1.0/en/0x10-C10-MCP-Security.md

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

AIUC1-Q3-2026**AIUC-1 Q3 2026 standard update**

Artificial Intelligence Underwriting Company / assurance standard / Q3 2026 / 2026-07-14 / Accessed 2026-09-19

Coding-agent controls for credentials, generated-code security, containment, and responsibility boundaries. **Limit:** Privately maintained assurance program. Scanner output is not certification or completion of independent evaluation.

www.aiuc-1.com/research/2026-q3-standard-update

ISO-42001-2023**ISO/IEC 42001:2023 - AI management systems**

ISO/IEC / management system standard / Edition 1 / 2023-12 / Accessed 2026-09-19

Public description of an organizational AI management system. **Limit:** Public description only; proprietary standard text not reproduced and no clause-level compliance mapping.

www.iso.org/standard/42001

ISO-23894-2023**ISO/IEC 23894:2023 - AI guidance on risk management**

ISO/IEC / risk management guidance / Edition 1 / 2023-02 / Accessed 2026-09-19

Public description of integrating AI risk management into organizational activities. **Limit:** Public abstract only; proprietary clauses not reviewed or mapped.

www.iso.org/standard/77304.html

OPENSSEF-BASELINE-202608**Open Source Project Security Baseline**

OpenSSF / project security baseline / v2026.08.28 / 2026-08-28 / Accessed 2026-09-19

Project-maturity baseline for source, builds, releases, documentation, and vulnerability handling. **Limit:** General project security, not agent behavior or deployed authorization assurance.

baseline.openssf.org/versions/2026-08-28

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

SLSA-12

Supply-chain Levels for Software Artifacts specification

SLSA / OpenSSF / supply chain specification / 1.2, approved / Release date not stated on reviewed specification page / Accessed 2026-09-19

Source and build integrity, provenance, attestations, and verification. **Limit:** Artifact origin and build integrity do not establish safe agent behavior or absence of vulnerabilities.

slsa.dev/spec/v1.2/

OPENSSEF-SCORECARD

OpenSSF Scorecard

OpenSSF / repository assessment tool / Living project / Living source; reviewed 2026-09-19 / Accessed 2026-09-19

Automated repository and dependency security signals. **Limit:** Heuristic scores do not certify security; not executed for this publication.

securityscorecards.dev/

OPENSSEF-MODEL-SIGNING

Launch of Model Signing v1.0

OpenSSF / artifact integrity tooling / v1.0 launch reference / 2025-04-04 / Accessed 2026-09-19

Signing and verifying model-artifact origin and integrity. **Limit:** A signature is not evidence that a model is free from backdoors, unsafe behavior, or licensing concerns.

[openssf.org/blog/2025/04/04/launch-of-model-signing-v1-0-openssf-ai-ml-working-group-secures-the...](https://openssf.org/blog/2025/04/04/launch-of-model-signing-v1-0-openssf-ai-ml-working-group-secures-the-...)

BENCH-MSB

MCP Security Bench (MSB)

Dongsen Zhang and coauthors / executable research benchmark / Paper v2; ICLR 2026; living repository / 2026-03-24 / Accessed 2026-09-19

Real MCP tool attacks across planning, invocation, and response handling with a security/utility trade-off metric. **Limit:** Distinct from MCPSecBench; selected tools, domains, agents, and code revisions bound results. Not executed here.

github.com/dongsenzhang/MSB

07 / SOURCE DIRECTORY

Follow the evidence

Primary-source links, version context and the recorded access date for each source.

BENCH-MCP-SAFETY

MCP-SafetyBench

Xuanjun Zong and coauthors / executable research benchmark / Paper v2; living repository / 2026-03-05 / Accessed 2026-09-19

Multi-turn, cross-server safety evaluation using real MCP servers in five domains and 20 attack types. **Limit:** Service state, harness assumptions, and task selection affect reproducibility; not protocol authorization certification. Not executed here.

github.com/xjzzzzzzzz/MCPSafety

BENCH-AGENTDYN

AgentDyn

SaFo-Lab / Hao Li and coauthors / executable research benchmark / Paper v3; living repository / 2026-05-07 / Accessed 2026-09-19

Dynamic open-ended tasks and helpful third-party instructions expose over-blocking by injection defenses. **Limit:** Extends AgentDojo and inherits harness assumptions; avoid double-counting identical scenarios. Not executed here.

github.com/SaFo-Lab/AgentDyn

BENCH-AGENTHARM

AgentHarm

Gray Swan AI / UK AI Safety Institute and coauthors / executable research benchmark / ICLR 2025; living Inspect Evals implementation / 2025 / Accessed 2026-09-19

Harmful multi-step user requests and jailbreaks with corresponding benign tasks. **Limit:** Different threat model from indirect injection. Grading can use LLMs; public subsets, scoring revisions, and safety/security-use license apply. Not executed here.

github.com/UKGovernmentBEIS/inspect_evals/blob/main/src/inspect_evals/agentharm/README.md

BENCH-BIPIA

BIPIA

Microsoft / Jingwei Yi and coauthors / executable research benchmark / 2023 initial paper; living repository / 2023 / Accessed 2026-09-19

Indirect prompt injection in external email, web, table, summarization, and code content. **Limit:** External-content robustness, not full agent/MCP transport security; some datasets require separate licensed acquisition. Not executed here.

github.com/microsoft/BIPIA

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TECH-GITHUB-ACTIONS**GitHub Actions: Secure use reference**

GitHub / Vendor security documentation / Living GitHub Actions documentation / Living documentation / Accessed 2026-09-19

Third-party workflow action integrity, full commit pinning, workflow permissions, secrets and untrusted input handling. **Limit:** The older cited URL redirects to the current Secure use pinned reference. Pinning mitigates mutable-action changes but does not prove the pinned action is safe.

docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-gi...

TECH-PYTHON-SECURITY**Python Security Considerations**

Python Software Foundation / Language security documentation / Python 3.14.7 documentation at review / Living documentation / Accessed 2026-09-19

Security-sensitive standard-library APIs, including serialization, subprocesses, temporary files and related trust boundaries. **Limit:** The /3 URL tracks current Python documentation. Check the installed runtime and follow linked module-specific guidance; this overview is not an agent/MCP security standard.

docs.python.org/3/library/security_warnings.html

TECH-PYTHON-TAR**Python tarfile: Extraction filters**

Python Software Foundation / Language security documentation / Python 3.14.7 documentation at review / Living documentation / Accessed 2026-09-19

Extraction filters, path and link restrictions, trusted archives and additional resource controls. **Limit:** Default extraction behavior differs by Python version and security backports. A data filter does not prevent every denial-of-service or unsafe-file outcome; inspect inputs and constrain resource use.

docs.python.org/3/library/tarfile.html#extraction-filters

TECH-NODE-PROCESS**Node.js Child process**

Node.js contributors / Runtime API documentation / Node.js v26.9.0 documentation at review / Living documentation / Accessed 2026-09-19

Shell interpretation and process-spawning behavior, including risks from untrusted input passed to exec or shell-enabled spawn. **Limit:** The URL follows current Node documentation; platform and runtime versions matter. Static shell-call detection does not establish attacker-controlled data flow.

nodejs.org/api/child_process.html

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Follow the evidence

Primary-source links, version context and the recorded access date for each source.

TECH-PYTORCH-SERIAL

PyTorch Serialization semantics

PyTorch contributors / Framework security documentation / PyTorch 2.14 documentation at review / Living documentation / Accessed 2026-09-19

Checkpoint trust, torch.load, restricted weights-only loading and serialization security limits. **Limit:** The old domain redirects to docs.pytorch.org and its stable page resolves to 2.14. Defaults vary by PyTorch version; restricted loading is not a universal defense against malicious artifacts.

pytorch.org/docs/stable/notes/serialization.html

TECH-PYYAML

PyYAML Documentation

PyYAML maintainers / Library documentation / Living documentation; includes historical examples / Living documentation / Accessed 2026-09-19

YAML object construction, SafeLoader and safe_load behavior when processing untrusted documents. **Limit:** Examples span historical PyYAML versions; assess the installed loader and registered constructors. Safe loading still needs application schemas, input limits and resource controls.

pyyaml.org/wiki/PyYAMLDocumentation

CISA-SECURE-BY-DESIGN

Shifting the Balance of Cybersecurity Risk: Principles and Approaches for Secure by Design Software

CISA and U.S./international partners / Government secure-development guidance / October 2023 update / 2023-10-25 / Accessed 2026-09-19

Manufacturer responsibility for security outcomes, accountable development and secure product defaults. **Limit:** General engineering rationale for several scanner rules. The publication does not define these individual source patterns, severity ratings or a pass/fail agent security benchmark.

www.cisa.gov/resources-tools/resources/secure-by-design

AGENT-SKILLS-SPEC

Agent Skills specification

Agent Skills maintainers / Format specification / Living specification / Undated living specification / Accessed 2026-09-19

SKILL.md metadata, Markdown instructions, optional scripts/references/assets and experimental allowed-tools declarations. **Limit:** Defines the skill format, not a security certification or maliciousness benchmark. Threat predicates and remediation are authored by Invarune; actual client permission enforcement varies.

agentskills.io/specification

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Follow the evidence

Primary-source links, version context and the recorded access date for each source.

TECH-PYTHON-CHMOD

Python os.chmod and permission mode flags

Python Software Foundation / API documentation / Python 3 living documentation / Undated living documentation / Accessed 2026-09-23

Permission bits applied by os.chmod, fchmod and lchmod; platform limitations. **Limit:** Documents API semantics, not proof of runtime access or an AI security certification. Windows and ACL behavior require deployment validation.

docs.python.org/3/library/os.html#os.chmod

CWE-732

CWE-732: Incorrect Permission Assignment for Critical Resource

MITRE / Weakness taxonomy / Living CWE definition / Undated living definition / Accessed 2026-09-23

Incorrect permissions allowing unintended access to a critical resource. **Limit:** A weakness classification, not an exploitability finding or complete permission audit. Agent/MCP mappings are authored by Invarune.

cwe.mitre.org/data/definitions/732.html
